Expert Systems With Applications 334 (2027) 134443



Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa



TriSV-CD: Selective verification with a do-no-harm arbitration gate for reliable remote sensing change detection ⋆

Zhenyu Wang, Gang Shi *

School of Computer Science and Technology, Xinjiang University, No. 777 Huarui Street, Shuimogou District, Urumqi, 830017, Xinjiang, China

ARTICLE INFO

Keywords:
Change detection
Reliable verification
Selective prediction
Vision-language model
Do-no-harm arbitration
Remote sensing

ABSTRACT

Vision–language models (VLMs) can diagnose disputed regions in remote sensing change detection (CD), but placing their prompt-sensitive decisions on an automatic correction path creates a reliability risk. We audit this behavior with Qwen2.5-VL models and find that direct accept/rollback outputs vary with prompt wording, while descriptive labels remain useful for error analysis but also vary with context and model scale. We therefore separate offline VLM diagnosis from deployment-time arbitration and propose **V4-Safe**, a training-free rollback gate for a test-time-augmentation correction proposal. V4-Safe measures the baseline probability on the proposal's edit support and restores the baseline when that support lies in a high-confidence region; it requires no VLM inference at deployment. Across a complete five-backbone by three-dataset matrix, V4-Safe yields positive mean F1 changes in 13 cells, an exact no-op in one degenerate cell, and a statistically indistinguishable −0.000001 change in one cell; 9 of 15 gains remain significant after Holm correction. It reduces harmful edit rate from 41.90% to 7.68% on PDEFM-SYSU and from 10.87% to 4.88% on BIT-NJDS. External-backbone and zero-shot diagnostics further expose probability-scale and domain-transfer limits of the fixed threshold. These results position V4-Safe as an empirical reliability-control layer rather than a universal safety guarantee, and show why CD verification should report harmful edits, uncertainty intervals, and rollback-or-accept Oracle regret in addition to mean F1.

1. Introduction

Remote sensing change detection (CD) aims to identify surface changes between two co-registered images acquired over the same geographic area at different times. It underpins a wide range of downstream applications, including urban growth monitoring, infrastructure updating, land-use analysis, and post-disaster assessment (Chen & Shi, 2020; Shi et al., 2022). Deep learning has substantially advanced CD performance over the past decade, with Siamese convolutional networks (Daudt et al., 2018; Fang et al., 2022), transformer-based architectures (Bandara & Patel, 2022; Chen et al., 2022), and more recent hybrid or state-space designs (Gao et al., 2024; Zhang et al., 2025) pushing the mean F1 score on standard benchmarks to increasingly high levels.

Despite this progress, the *reliability* of CD predictions has received much less attention than raw accuracy. In high-stakes applications such as damage assessment or infrastructure auditing, a silent false alarm or a missed change can have disproportionate downstream consequences, yet the dominant evaluation protocol, mean per-image F1 across a benchmark split, remains insensitive to the *distribution* of errors. Two CD systems with identical mean F1 may disagree sharply on which samples they fail on, how confidently they fail, and whether a post-hoc verification step would help or hurt.

A natural response has been to augment CD pipelines with an additional *verification* or *correction* step that can override or refine the baseline prediction. Classical options include test-time augmentation (TTA) and selective prediction (Geifman & El-Yaniv, 2017; Wang et al., 2021). More recently, the rapid progress of vision–language models (VLMs) (Bai et al., 2025; Li et al., 2024) has motivated a new line of work that proposes to let a VLM *directly decide* whether a given change-map prediction should be accepted, suppressed, or modified (Liu et al., 2024a). The appeal is obvious: VLMs appear to offer high-level semantic reasoning that pixel-level networks lack.

However, whether current deployable VLMs are in fact *reliable enough* to control decisions in a CD pipeline has not been systematically examined. Most prior works either treat the VLM as one component of a larger system without isolating its contribution, or validate it only on a small curated set of samples. In this paper, we ask a sharper question:

> *Can a current open-source VLM be trusted as a direct verifier or correction decision-maker in CD, and if not, what role should it play in a reliable expert system?*

⋆ This work was supported by the Key Research and Development Project in Xinjiang Uygur Autonomous Region (No. 2022B01006).
* Corresponding author.
E-mail addresses: wangzhenyu1127@stu.xju.edu.cn (Z. Wang), shigang@xju.edu.cn (G. Shi).

https://doi.org/10.1016/j.eswa.2026.134443
Received 8 May 2026; Received in revised form 19 July 2026; Accepted 15 September 2026
Available online 16 September 2026
0957-4174/© 2026 Published by Elsevier Ltd.

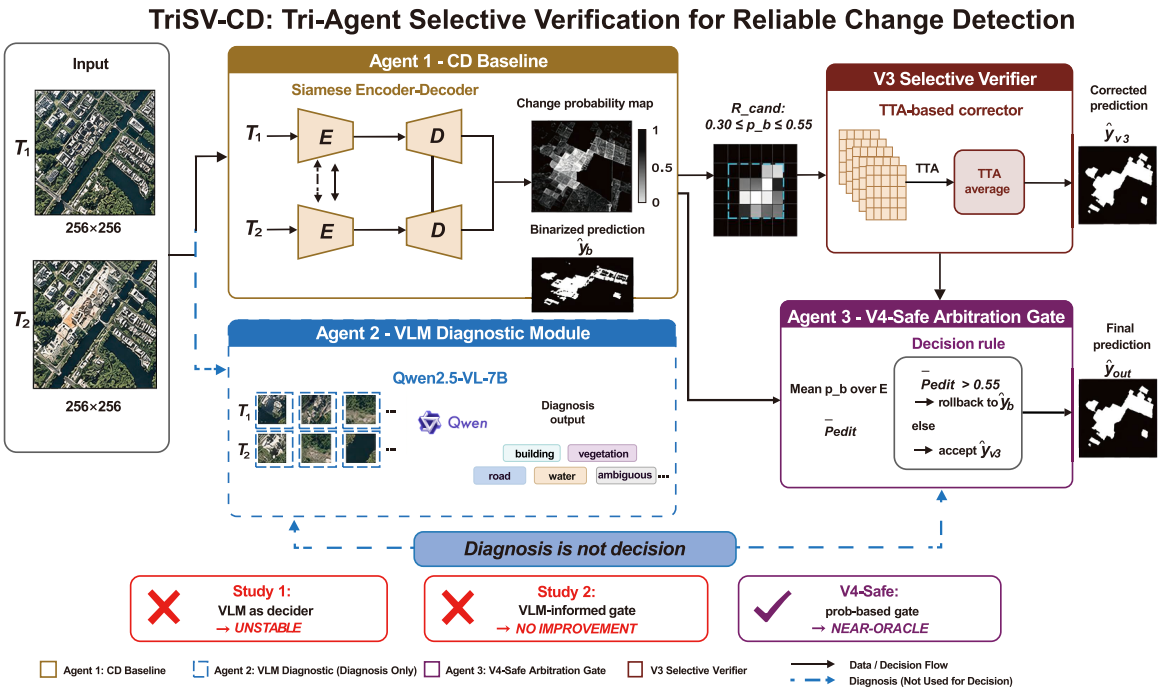


Fig. 1. Overall framework of TriSV-CD. The baseline CD model produces a probability map p_b and a binarized prediction $\hat{y}_b$. The uncertainty band $0.30 \le p_b \le 0.55$ is used only to identify disputed samples and crop candidate regions for V3/VLM diagnosis. V3 is a TTA-based proposal generator that outputs $\hat{y}_{v3}$. V4-Safe then computes $\bar{p}_{\text{edit}}$ on the candidate edit support $\mathcal{E} = \{(i,j) : \hat{y}_{v3}(i,j) \neq \hat{y}_b(i,j)\}$ and rolls back to the baseline when this support lies in a high-confidence baseline region. The VLM branch provides semantic diagnosis only and is intentionally disconnected from the final gate decision.

1.1. A two-study empirical investigation

We address this question with two controlled diagnostic studies.

Study 1: Five-prompt analysis of VLMs as direct verifiers. We instrument Qwen2.5-VL-7B with five prompt variants covering binary, ternary, scoring, and descriptive modes, and evaluate each on disputed correction candidates from a TTA-based selective verifier. The resulting decision distributions are highly unstable across prompts: different prompts produce suppression rates ranging from 0% to 100%. Open-ended change-type descriptions provide structured categorical outputs (building / road / vegetation / water / ambiguous) that are more useful for auditing than direct actions, although later context- and scale-sensitivity checks show that these labels are not invariant.

Study 2: VLM-informed gating at scale. One could nevertheless hope that even if VLMs cannot take a raw decision, their structured *diagnostic* output (the change-type label) might be combined with a hand-crafted rule to form a more reliable gate. We test this hypothesis directly. We run the VLM over the full SYSU test set (91.88% coverage on disputed samples, N=4,000) and assign per-category probability thresholds informed by domain knowledge (e.g., more conservative thresholds for vegetation, where seasonal pseudo-change is common). The resulting VLM-informed gate is slightly lower than a single fixed-threshold rule without VLM input, and the difference is not statistically significant: $\Delta F1 = -0.0004$, Wilcoxon $p = .672$. The semantic signal provided by the VLM, though interpretable, does not provide measurable additional decision value in this tested gate beyond what is already implicit in the baseline probability.

1.2. From empirical findings to a design principle

These two studies converge on a single conclusion for the tested open-source VLM and prompt families: in CD verification, semantic diagnosis is safer than automatic decision control. We codify this as the **decision–diagnosis separation principle**: *let the VLM describe; let a transparent, do-no-harm arbitration rule decide.*

Guided by this principle, we propose **V4-Safe**, a lightweight, training-free rollback gate for selective CD verification. V4-Safe is built on two observations that we establish empirically. First, the harmful edits of an underlying selective verifier (V3, a TTA-based corrector) are *not uniformly distributed*; they concentrate sharply in high-confidence regions of the baseline's output distribution (on PDEFM-SYSU, 95.45% of V3 harmful samples have $\bar{p}_{\text{edit}}$>0.55 versus 61.81% for beneficial samples, Mann–Whitney U p=8.7×10^{-31}). Second, the same cutoff is informative beyond SADE-SYSU, although its harmful-rate advantage is weaker on public backbones whose probability distributions are calibrated differently. V4-Safe therefore rolls back an edit whenever the baseline is already confident, on the principle that a verifier should *do no harm* to strong baseline predictions.

1.3. Reliability beyond mean F1

We argue that evaluating a selective verifier by mean F1 alone is fundamentally insufficient. A verifier that gains +0.005 mean F1 while 30% of its accepted edits are harmful is qualitatively different from one with a 7% harmful-edit rate. We therefore complement mean F1 with three reliability-oriented metrics: *harmful edit rate* (the fraction of retained edits whose per-sample F1 decreases by more than 0.005), *Oracle gap* (the distance from an oracle that selects the per-sample maximum of baseline and corrected F1, i.e., an upper bound on what any gate can achieve), and a *stratified reliability analysis* that partitions samples by baseline confidence and reports per-stratum behavior. These metrics show that V4-Safe improves average performance while reducing harmful edits in several cells, with reductions up to 5.5× (PDEFM-SYSU: 41.90%→7.68%) and operates at an Oracle gap at most 0.002 for the correction-active in-house cells and below 0.007 for the two external SYSU-CD cells, indicating small absolute distance to the sample-wise rollback-or-accept upper bound.

1.4. Contributions

Our contributions are as follows.

- **A controlled diagnostic study of VLM roles in CD verification.** We evaluate five prompt families and show that the tested open-source VLM is not reliable as an automatic accept/rollback decider under our prompts, while its descriptive change-type labels are useful for semantic auditing.
- **A full-test-set test of hand-crafted VLM-informed gating.** On SYSU-CD, a VLM-informed per-category gate with 91.88% coverage is slightly lower than a probability-only gate and is not significantly different from it (ΔF1=−0.0004, p=0.672). This supports the *decision–diagnosis separation* principle without claiming that all possible VLM integrations are ineffective.
- **V4-Safe, an empirically grounded rollback gate.** We analyze the harmful-edit landscape of a TTA-based correction proposal and show that many harmful edits occur where the baseline is already confident on the candidate edit support. V4-Safe rolls back such proposals using one transparent threshold and requires no VLM inference at deployment.
- **A reliability-oriented evaluation framework** built on harmful edit rate, Oracle gap, and stratified failure analysis. These metrics expose safety properties that mean-F1 reporting alone can hide.
- **Validation with explicit scope.** We evaluate a complete five-backbone by three-dataset correction matrix, with SADE-NJDS reported as a degenerate all-background/no-edit cell rather than removed. V4-Safe gives positive gains in 13 cells, an exact no-op in the degenerate cell, and a statistically indistinguishable −0.000001 change in SNUNet-LEVIR, with significant gains in 9 of 15 cells. We further add BIT/SNUNet SYSU-to-LEVIR/NJDS zero-shot diagnostics; the zero-shot cross-domain results are weak, so we report them as robustness evidence and limit the generalization claim rather than presenting a universal-success result.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 presents our two diagnostic studies on VLMs as CD verifiers. Section 4 formalizes V4-Safe. Section 5 reports the comprehensive empirical evaluation. Section 6 discusses limitations and design implications, and Section 7 concludes.

2. Related work

2.1. Deep learning for change detection

Early deep CD methods were dominated by fully convolutional Siamese networks. Daudt et al. (2018) introduced FC-Siam-conc and FC-Siam-diff, demonstrating the effectiveness of weight-shared encoders for bi-temporal analysis. Subsequent works such as SNUNet-CD (Fang et al., 2022) adopted dense connections for improved feature reuse. With the rise of attention and transformers, BIT (Chen et al., 2022) mapped deep features into tokens to model long-range semantic relations, and ChangeFormer (Bandara & Patel, 2022) adopted a hierarchical transformer encoder. More recent methods such as RCTNet (Gao et al., 2024) and CDMamba (Zhang et al., 2025) further improve performance through CNN-transformer fusion and state-space modeling; in particular, CDMamba adapts state-space modeling to capture long-range dependencies in bi-temporal features. Recent ESWA work has also advanced CD with progressive full-scale aggregation (Fu et al., 2026), hierarchical window aggregation (Yang et al., 2026), and interference-resistant dual-branch interaction (Yuan & Du, 2026). Attention mechanisms developed in adjacent dense-prediction fields, including medical image segmentation, are valuable architectural analogues for suppressing background clutter and preserving fine structure. We do not use such single-image medical segmentation networks as direct baselines because their input semantics, output targets, and supervision differ from paired remote-sensing CD. Instead, we compare against CD backbones that instantiate analogous long-range, channel/spatial-attention, aggregation, and transformer ideas under the paired bi-temporal setting. Beyond architectural innovation, recent work has also explored prompt-guided segmentation (Chen et al., 2024) and remote-sensing foundation models such as RemoteCLIP (Liu et al., 2024b), indicating a broader shift toward reusable visual representations. Multi-task frameworks such as ChangeMask (Zheng et al., 2022) offer richer supervision by jointly modeling change and semantic structure, but they increase annotation requirements and do not directly address post-hoc reliability.

Foundation-model adaptation has also entered remote-sensing CD. SAM-based studies range from zero-/one-shot remote-sensing segmentation (Osco et al., 2023) to parameter-efficient CD with feature interaction (Zhang et al., 2024a), limited-data adaptation (Gao et al., 2025), and EfficientSAM-assisted feature interaction with prior guidance (Peng et al., 2025). In parallel, cross-resolution CD has used Fourier and wavelet structure to model unequal bi-temporal relationships under resolution mismatch (Ning et al., 2025). These methods improve representation learning, boundary recovery, data efficiency, or resolution robustness. TriSV-CD addresses a different stage: after a trained backbone has produced a baseline probability map and a TTA correction proposal, it asks whether the proposed edit should be accepted or rolled back. The two directions are therefore complementary rather than competing architectural replacements.

Our three in-house baselines—SADE-Net (Wang et al., 2026b), PDEFM-Net (Wang et al., 2026a), and DCCA-Net (Wang et al., 2026c)—were designed to span the CNN-only, recurrent-augmented, and attention-augmented branches of this family tree, and the two public baselines (BIT, SNUNet) extend coverage to transformer and dense-connection paradigms. Crucially, our contribution is orthogonal to architectural innovation: V4-Safe is a lightweight *post-hoc* verification mechanism for a trained CD model that exposes a per-pixel probability map and the paired TTA correction proposal required to define the edit support. Aggregate benchmark scores or a final binary mask alone are insufficient to evaluate the rollback decision.

2.2. Reliability, calibration, and selective prediction

The general machine learning literature has extensively explored reliable prediction under distribution shift. Guo et al. (2017) documented the miscalibration of modern neural networks, while selective classification (Geifman & El-Yaniv, 2017; Jiang et al., 2018) formalized the principle of abstaining on low-confidence inputs. In change detection (CD), however, reliability-aware evaluation remains limited. Most CD papers report only mean precision, recall, F1, and IoU. These averages summarize error magnitude but obscure how errors are distributed across samples. More recently, conformal prediction (Angelopoulos & Bates, 2023) has been introduced as a distribution-free framework for coverage guarantees, with extensions to image-to-image prediction (Angelopoulos et al., 2022). Uncertainty estimation in remote sensing, while less mature than in medical imaging, has gained increasing attention. The survey of Gawlikowski et al. (2023) notes that calibration-aware evaluation remains uncommon in many dense prediction benchmarks. Deep ensembles offer a practical route to calibrated uncertainty without explicit Bayesian inference (Lakshminarayanan et al., 2017), while Ovadia et al. (2019) show that calibration degrades under distribution shift—an issue directly relevant to CD, where temporal gaps between acquisitions naturally induce such shifts. These reliability tools motivate our use of paired per-sample tests, harmful edit rate, and Oracle gap. However, our setting differs from standard selective prediction because the system must arbitrate between a baseline prediction and a post-hoc correction proposal. Our work draws on selective-prediction principles but inverts the operating regime: instead of abstaining on low-confidence inputs, V4-Safe *rolls back* edits in the *high-confidence* region, based on the empirical

observation that harmful edits from TTA-based correction concentrate there (Section 4.2).

2.3. Post-hoc arbitration as an expert-system layer

From an expert-systems perspective, a verification module should not merely raise an average score; it should encode operational rules that are inspectable, auditable, and conservative under uncertainty. This requirement is particularly important in CD, where a final map may support downstream human decisions in urban planning or disaster response. Classical selective classification focuses on rejecting uncertain inputs, whereas the present setting is a two-candidate arbitration problem: the system must choose between a baseline prediction and a proposed correction. This distinction motivates our "do-no-harm" formulation. The gate is not intended to replace a CD backbone; rather, it is a transparent rule layer that can be inspected through harmful rate, Oracle gap, and stratum-level failure profiles. This positioning also explains why a simple rule can be a meaningful contribution for ESWA: its value lies in reliable decision control and diagnostic interpretability, not in adding a heavier segmentation architecture.

2.4. Test-time augmentation and post-hoc correction

Test-time augmentation and fully test-time adaptation (Wang et al., 2021) have been widely used to robustify predictions without retraining. Applied to CD, TTA typically averages predictions over several geometric transforms to reduce variance near decision boundaries. Our selective verifier, V3, is a minimal TTA-based corrector that proposes edits for disputed samples; it forms the *input* to the V4-Safe gate. Importantly, we do not claim V3 as a contribution; rather, we use V3 as a representative, deliberately imperfect correction proposal, and study how to arbitrate its outputs reliably. Several variants of test-time adaptation have been proposed beyond simple geometric augmentation. Test-time training (TTT) updates model parameters at test time using a self-supervised auxiliary task (Sun et al., 2020), whereas MEMO adapts a single sample by enforcing consistency across augmented views (Zhang et al., 2022). These methods improve robustness but often assume that adaptation or augmentation is beneficial when applied. In dense prediction, the cost of a harmful correction can be spatially localized but practically important: a small number of wrongly edited patches may dominate downstream interpretation. V4-Safe therefore treats TTA as a proposal generator and adds a rollback layer rather than accepting all augmented predictions without a safety check. The core question is not "can TTA correct?" but "given that a corrector will sometimes harm, how should a gate decide when to accept it?"

2.5. Vision-language models in remote sensing

The emergence of large-scale vision–language models (Bai et al., 2025) has motivated increasing research on remote-sensing image captioning, object description, and, more recently, interactive change analysis (Li et al., 2024; Liu et al., 2024a). Recent RS-specific VLMs cover several related capabilities. RSGPT (Hu et al., 2025) adapts language-model-style generation to remote-sensing image captioning, whereas GeoChat (Kuckreja et al., 2024) supports grounded multi-turn dialogue for scene understanding. EarthGPT (Zhang et al., 2024b) focuses on multi-sensor interpretation, and RemoteCLIP adapts contrastive vision-language pretraining to remote-sensing imagery. These models show that VLMs can provide useful semantic descriptions, but they do not by themselves establish that VLM outputs are stable enough to drive correction decisions in CD. Several works propose VLMs as direct verifiers or correctors inside CD pipelines. However, to the best of our knowledge, no prior study systematically quantifies the *stability* of VLM-driven correction decisions across prompt variants, nor tests whether a hand-crafted VLM-informed gating mechanism can improve over a simple probability rule at scale. This is precisely the gap addressed by our two diagnostic studies in Section 3.

2.6. Positioning

In summary, our work sits at the intersection of three strands: architecturally informed CD (from which we inherit baselines), selective/reliable prediction (from which we inherit the evaluation philosophy), and VLM-augmented remote sensing (which motivates our diagnostic studies). Rather than proposing a heavier CD architecture or a new VLM-based pipeline, we argue that *simpler, empirically grounded arbitration* can deliver reliability gains while avoiding the prompt-sensitivity risks observed in the tested VLM-as-decider setting.

3. Diagnostic studies of VLMs as CD verifiers

This section presents two controlled empirical studies that jointly characterize the role VLMs can reliably play in a CD verification pipeline. Study 1 (Section 3.2) evaluates the tested VLM as a direct decision-maker under five prompt variants. Study 2 (Section 3.3) tests whether the structured change-type output can inform a hand-crafted gating rule at scale. Both studies converge on the *decision–diagnosis separation* principle that motivates our V4-Safe gate (Section 4).

3.1. Setup: disputed samples and the V3 selective verifier

We first define the setting common to both studies. Given a CD baseline producing per-pixel change probabilities on a bi-temporal image pair, we identify *disputed regions* as contiguous areas where the baseline probability falls in an uncertainty band $[\tau_{\text{low}}, \tau_{\text{high}}]=[0.30, 0.55]$. A TTA-based corrector, denoted **V3**, produces an alternative prediction for each disputed sample as follows. We apply eight geometric augmentations (horizontal flip combined with $\{0°, 90°, 180°, 270°\}$ rotations) to both input images, perform a forward pass through the *same* trained baseline model for each augmentation, and average the resulting probability maps over $\mathcal{R}_{\text{cand}}$ to produce a denoised probability estimate. This estimate is binarized at 0.5 to yield the V3 prediction $\hat{y}_{\text{v3}}$. No auxiliary corrector network is trained; V3 is purely a TTA-based re-inference of the original baseline (the corrector block in Fig. 1 is a simplified illustration of this averaging process). V3 is intentionally simple, and its harmful-edit rate is non-trivial—especially on non-SADE architectures (Section 5.2)—which makes it an informative input to any subsequent gate.

For each disputed sample we therefore have three predictions: baseline $\hat{y}_{\text{b}}$, V3 correction $\hat{y}_{\text{v3}}$, and ground truth y. The downstream arbitrator observes the candidate edit support $\mathcal{E} = \{(i,j) : \hat{y}_{\text{v3}}(i,j) \neq \hat{y}_{\text{b}}(i,j)\}$ and the scalar $\bar{p}_{\text{edit}}$, defined in Section 4.1 as the mean baseline probability over this support. This distinction is important: the uncertainty band mines a disputed sample, whereas $\mathcal{E}$ describes the pixels V3 actually proposes to change. Any downstream arbitrator must then decide whether to accept $\hat{y}_{\text{v3}}$ ("*accept*") or roll back to $\hat{y}_{\text{b}}$ ("*rollback*").

3.2. Study 1: VLMs as direct verifiers are unstable across prompts

We instrument Qwen2.5-VL-7B-Instruct (Bai et al., 2025) with five prompt variants that cover the decision space a VLM verifier might occupy. V1–V3 receive the same four aligned panels (T1, T2, baseline change map, and V3 change map). V4 is an earlier three-panel confidence probe and is retained as an auxiliary failure example; the clean paired prompt-sensitivity comparison is therefore V2 versus V3 on the same 532 patches. V5 uses the three-image open-description input (T1, T2, predicted change mask) specified in Appendix B.

- **V1 (binary suppress, N=1000).** "Should the proposed edit be *kept* or *suppressed*?"
- **V2 (binary keep, N=100).** Same as V1 with reversed wording to test prompt symmetry.

Table 1
VLM prompt variants and their decision distributions on SADE-SYSU disputed samples. V1–V4 are pilot prompt probes with different sample sizes and exhibit severe prompt sensitivity; V5 (descriptive) is later run at full scale.

Variant	N	Task Mode	Dominant Output
V1	1000	Binary (suppress/keep)	Suppress ∼50%
V2	100	Binary (keep/suppress)	Keep ∼37%
V3-prompt	100	Ternary (+uncertain)	Uncertain ≈100%
V4-prompt	30	Confidence scoring	Suppress ∼100%
V5	4000	Change-type description	Structured categories

Table 2
VLM-informed per-category gate vs. vanilla V4-Safe on SADE-SYSU (N=4,000). Per-category thresholds: building τ=0.50, road 0.55, vegetation 0.60, water 0.60, ambiguous 0.55. The VLM-informed gate does not outperform vanilla.

Method	Mean F1	ΔF1	Harmful (%)
Baseline	0.6563	—	—
Vanilla V4-Safe	0.6614	+0.0051	18.36
VLM-informed gate	0.6610	+0.0047	19.67
Difference (VLM vs vanilla): ΔF1=−0.0004, p=0.672			

- **V3-prompt (ternary with uncertainty, N=100).** "Keep / Suppress / Uncertain."
- **V4-prompt (confidence scoring, N=30).** "Score the reliability of the proposed edit $\in [0, 1]$."
- **V5 (descriptive, N=30, and later N=4000 in Study 2).** "Describe the dominant change type in this bi-temporal pair (building / road / vegetation / water / other/ambiguous)."

Prompt input and parsing. For every prompt run, all panels belonging to a patch use the same integer window coordinates in the original 256×256 sample. The base windows are square 64×64 tiles on a 64-pixel grid, so no aspect-ratio-changing resize or independent recropping is applied between T1, T2, prediction, probability, ground truth, and edit-support arrays. The VLM receives the native square RGB PNGs through the Qwen processor; any processor-internal tokenization is separate from the original-resolution V4-Safe computation. V4-Safe never maps a VLM crop back to the output mask: its statistic is computed directly from the full-resolution baseline probability and edit-support arrays.

Outputs are parsed with fixed string rules ("keep", "suppress", "uncertain", or one of the declared change-type labels). Invalid or unparsed outputs are counted explicitly rather than silently reassigned. Appendix B reports the prompt instructions, decoding settings, input sizes, call counts, and a paired example.

Findings. The results are summarized in Table 1.

V1 suppresses ∼50% of edits. V2 (with reversed wording) keeps ∼37%, a gap that cannot be explained by any actual change in the underlying samples. V3-prompt returns *uncertain* on almost all sampled cases, effectively refusing to commit. V4-prompt collapses to nearly 100% suppression, regardless of input. These four decision-oriented prompts exhibit strong prompt sensitivity and inconsistent suppression behavior, making them unsuitable as decision-makers in a CD pipeline.

To rule out the possibility that this observation is only caused by unequal sample sizes, we also compute agreement on the overlapping patches that appear in multiple prompt runs. V2 and V3 share 532 patches and agree on 334 of them (62.78%), leaving a 37.22% switch rate. The intersection of V2, V3, and V4 contains 182 identical patch coordinates from 30 samples. On this auxiliary common-coordinate set, V2 outputs 85 keep and 97 uncertain decisions, V3 outputs 182 uncertain decisions, and V4 outputs 182 suppress decisions. The V2–V4 and V3–V4 agreement rates are therefore both 0% on the common set. This common-set audit confirms that the broad collapse pattern is not merely an artifact of different sample pools; the same-input V2–V3 comparison is the stricter prompt-only control.

V5 instead returns a structured change-type description that is useful for semantic auditing. We do not treat that description as invariant: the later context-size and 32B/72B checks show substantial label shifts.

Conclusion of Study 1. The tested open-source ∼7B VLM behaves differently as a *describer* versus a *decider*: structured descriptions are useful for auditing, whereas binary, ternary, or scalar reliability decisions are too prompt-sensitive for automatic control. Neither role is claimed to be invariant across context or model scale. A responsible CD system should therefore not place this kind of VLM directly on the automatic decision path without additional validation.

3.3. Study 2: can VLM-informed category thresholds do better than a single rule?

A natural next step is to use the VLM's structured output (change type) and use it to inform a rule that decides whether to accept or roll-back. A hand-crafted variant assigns different thresholds to different semantic classes and tests whether these labels add useful decision information beyond the baseline probability. We call this design the *VLM-informed per-category gate*.

Design. For every disputed sample we obtain its VLM change-type label via the V5 prompt. The label is one of five parsed classes: building, road, vegetation, water, or ambiguous. We then assign a per-category threshold τ_c by a fixed, non-optimized rule rather than by test-set tuning: $\tau_{\texttt{building}}$=0.50, $\tau_{\texttt{road}}$=0.55, $\tau_{\texttt{vegetation}}$=0.60, $\tau_{\texttt{water}}$=0.60, $\tau_{\texttt{ambiguous}}$=0.55. The gate rolls back an edit whenever $\bar{p}_{\text{edit}} > \tau_c$; otherwise it accepts V3's correction.

Scale and coverage. We run V5 over the entire SYSU test set, producing change-type labels for all samples with a non-trivial disputed region. The resulting coverage is 3,675/4,000 = 91.88% (the remaining 8.12% have no disputed region and therefore no VLM judgement is required). Within disputed samples, the VLM's change-type distribution is vegetation 52.11%, building 33.77%, road 7.02%, water 0.33%, ambiguous 6.72%. This is a coverage and distribution result under the fixed 7B/V5 setting, not evidence of invariance across prompt, context, or model scale.

Result. We then compare the VLM-informed gate to a *vanilla* single-threshold gate (τ=0.55 regardless of category), using SADE-SYSU as the baseline. Across N=4000 samples:

- Vanilla V4-Safe: mean F1 = 0.6614, ΔF1=+0.0051, harmful rate = 18.36%.
- VLM-informed gate: mean F1 = 0.6610, ΔF1=+0.0047, harmful rate = 19.67%.
- Difference vs vanilla: ΔF1=−0.0004, Wilcoxon signed-rank p = .672.

Table 2 and Fig. 2 present the full comparison.

Additionally, Table 3 reports the VLM change-type distribution on disputed samples.

To test whether this behavior is only an artifact of the 7B model size, we also ran two open-source scale checks using the same three-image open-description input. First, Qwen2.5-VL-32B-Instruct-AWQ was evaluated on 500 SYSU disputed samples (2395 patches). The 32B-AWQ model produced valid descriptions for all processed patches. At the patch level, the predicted semantic types were vegetation 71.65%, building 14.49%, road 11.61%, water 2.00%, demolition 0.13%, and shadow 0.13%. Confidence was high for 83.67%, medium for 16.28%, and low for 0.04% of patches. At the sample-majority level, however, the outputs remained concentrated on vegetation (78.8%) and high confidence

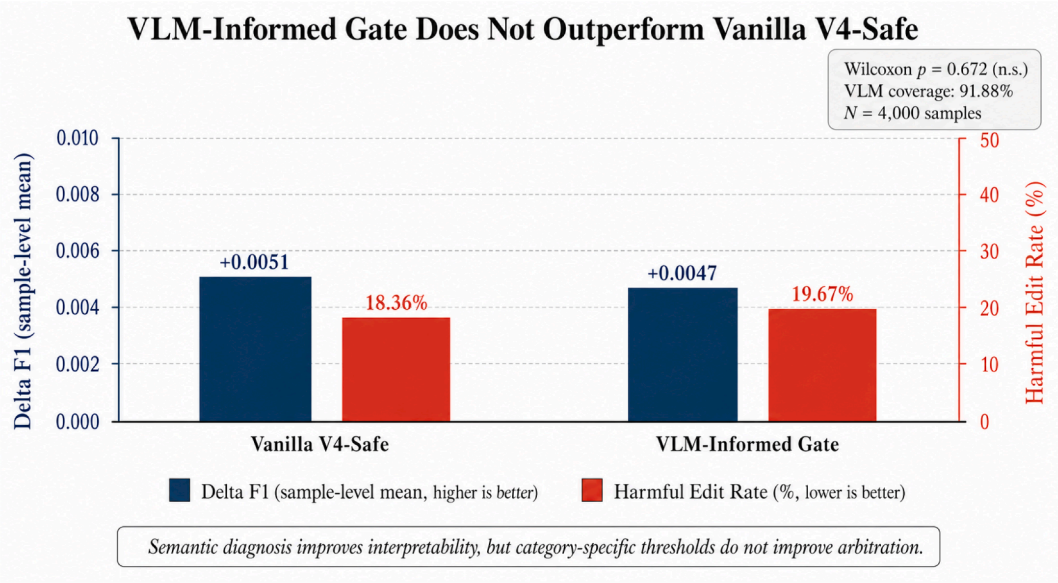


Fig. 2. VLM-informed per-category gate versus vanilla V4-Safe on SADE-SYSU. Despite 91.88% VLM coverage, the category-informed gate does not outperform the single-threshold rule (ΔF1=−0.0004, p=0.672).

Table 3
VLM V5 change-type distribution on SYSU-CD disputed samples (N=3,675, coverage 91.88%).

Change Type	Count	Proportion
Vegetation	1915	52.11%
Building	1241	33.77%
Road	258	7.02%
Ambiguous	247	6.72%
Water	12	0.33%
Other/unparsed	2	0.05%
No disputed region	325	(8.12% of 4,000)

Table 4
Change-type alignment on the exact common 100 samples. Rows are the 32B-AWQ sample-majority labels and columns are the 72B-AWQ labels. The diagonal sums to 55, matching the 55% agreement rate.

32B \ 72B	Building	Other	Road	Vegetation	Water
Building	0	2	2	1	0
Other	0	14	1	0	1
Road	0	5	2	0	0
Vegetation	5	14	9	39	1
Water	0	4	0	0	0

(88.2%). Applying the same category-threshold gate to this 500-sample subset gave an above-baseline result (ΔF1 = +0.00298), but did not improve over the vanilla probability-only V4-Safe gate on the same subset (vanilla V4-Safe: 0.73618; 32B-informed V4-Safe: 0.73539; difference: −0.00079), with harmful edit rate 22.22% and correction precision 59.14%.

Second, we downloaded and evaluated Qwen2.5-VL-72B-Instruct-AWQ. On an initial 100-sample SYSU disputed subset, the 72B-AWQ sample-majority distribution was vegetation 40.0%, other/ambiguous 39.0%, road 14.0%, building 5.0%, and water 2.0%; confidence was high for 92.0% and low for 8.0% of samples. On the same 100-sample subset, the 72B-informed gate achieved mean F1 0.70529 (ΔF1 = +0.01340 over baseline), slightly above the vanilla V4-Safe subset result of 0.70460 (difference +0.00069), with harmful edit rate 6.15% and correction precision 89.47%. We further compared the 32B and 72B outputs on their exact 100-sample overlap. The sample-level semantic labels agreed on 55 of 100 samples, while confidence levels agreed on 90 of 100 samples; the 72B model reduced the 32B vegetation-dominant tendency (68.0% vegetation for 32B vs. 40.0% for 72B) but increased ambiguous/other labels (16.0% vs. 39.0%) (Table 4).

We then expanded the 72B-AWQ run to 500 SYSU samples. The 500-sample sample-majority distribution was vegetation 59.6%, other 21.0%, road 14.4%, building 3.6%, and water 1.4%, with high confidence on 96.0% of samples and ambiguous samples in 18.4%. On this larger matched subset, the 72B-informed gate remained above baseline (mean ΔF1 = +0.00336), but did not improve over vanilla V4-Safe: mean F1 was 0.73578 for the 72B-informed gate versus 0.73618 for vanilla V4-Safe (difference −0.00040), with harmful edit rate 22.12% and correction precision 58.52%. Thus, the favorable 100-sample 72B result does not persist after expansion to 500 samples. These scale checks confirm that larger open-source VLMs can provide useful diagnostic descriptions and may improve selected subsets, but the semantic labels are not stable enough to serve as the main deployment rule. Therefore, the main deployed result remains the fully evaluated probability-only V4-Safe gate, because it is evaluated on the full test set and requires no online VLM inference.

Finally, we tested whether VLM semantic labels are sensitive to the crop context provided around each disputed patch. Using Qwen2.5-VL-32B-AWQ on the same 100 SYSU samples, increasing the visual context from the original 64-pixel disputed patches to 128 and 192 pixels substantially changed the sample-majority semantic labels. The 64-pixel view was vegetation-dominant (74%), the 128-pixel view shifted to building/vegetation/road (34%/29%/16%), and the 192-pixel view shifted further toward building/road (47%/27%). Semantic agreement was 32% between 64 and 128 context, 13% between 64 and 192 context, and 58% between 128 and 192 context. This context sensitivity supports using VLM outputs as diagnostic explanations rather than as the deployment-time correction authority.

The VLM-informed gate is individually better than the baseline (e.g. VLM-informed vs. baseline $p = 1.72\times10^{-24}$), but it does *not* improve over the vanilla gate. The direct comparison against vanilla yields a small negative difference and no statistical evidence of superiority (p = .672). A per-category breakdown reveals that no category class gains from VLM-informed thresholds: Δ vs. vanilla is −0.0004 (building),

0.000 (road), −0.0005 (vegetation), −0.0099 (water, N=12, dominated by small-sample noise).

We further test whether the category assignment itself carries stable decision value by randomly shuffling VLM labels 1000 times while keeping the same per-category thresholds. The original VLM category gate obtains ΔF1=+0.004658, harmful rate = 19.67%, and correction precision = 63.20%. The shuffled gates have mean ΔF1=+0.004830 (standard deviation 0.000121), with 5th/50th/95th percentiles +0.004627/ + 0.004831/ + 0.005026. In 92% of shuffle runs, the shuffled labels match or exceed the original category-gate delta. Thus, in this artifact set, the semantic category labels do not provide a stable deployable decision signal beyond probability.

As an additional fusion control, we train threshold rules on deployable probability features plus the 32B-AWQ sample-level semantic type on one half of the 500-sample common subset and evaluate on the held-out half. The best probability-only rule obtains held-out ΔF1 = +0.003352, harmful edit rate 21.83%, and correction precision 59.21%. The best semantic-type fusion rule obtains held-out ΔF1 = +0.003308, harmful edit rate 24.04%, and correction precision 57.63%, a −0.000044 held-out difference from the probability-only rule. Thus, adding VLM semantic type to deployable probability features does not improve held-out arbitration in this setting.

Interpretation. The result is unlikely to be explained solely by insufficient coverage (91.88%) or sample size (N=4000). A parsimonious interpretation is *redundancy*: the VLM's change-type signal does not provide measurable additional decision value *beyond* what is already implicit in the baseline probability. Baseline probability distributions already differ across change types (e.g., building edits have systematically different $\bar{p}_{\text{edit}}$ distributions from vegetation edits), and a single threshold applied after this implicit stratification is at least as good as hand-crafted per-category tuning.

Conclusion of Study 2. Even when used in structured description mode rather than direct decision mode, the tested VLM deployed as a gate-informer does not improve over a simple probability-based rule. Taken together, Studies 1 and 2 support the more cautious *decision–diagnosis separation* principle used in this paper.

3.4. Implication for method design

Studies 1 and 2 weaken two simple ways of putting the tested VLM on the decision path: direct verification (unstable across prompts) and hand-crafted VLM-informed gating (does not improve over a simple rule). They do *not* rule out the use of VLM outputs as diagnostic annotations, nor richer learned integrations of VLM features (which we leave to future work).

These findings motivate a probability-based gate for the main deployment path. The remaining question is: how should such a gate be designed to reliably reduce harmful edits without sacrificing mean F1? Section 4 formalizes V4-Safe, and crucially grounds its τ=0.55 cutoff in an empirical analysis of V3's harmful-edit landscape (Section 4.2).

4. V4-Safe: a do-no-harm rollback gate

Section 3 suggests that the tested VLM is better used for semantic diagnosis than for automatic accept/rollback decisions. We therefore separate diagnosis from decision and introduce **V4-Safe**, a lightweight post-hoc arbitration layer between a baseline CD model and the V3 TTA proposal. V4-Safe is intentionally simple: it does not introduce a new segmentation architecture, does not train a gating network, and does not require VLM inference at deployment. Its contribution lies in the reliability-oriented formulation of the rollback decision, together with an empirical audit of where such a simple rule succeeds or fails.

4.1. Formulation

Let $p_{\text{b}} \in [0,1]^{H\times W}$ be the baseline probability map and $\hat{y}_{\text{b}} \in \{0,1\}^{H\times W}$ its binarized prediction. V3 produces an alternative prediction $\hat{y}_{\text{v3}}$ by TTA averaging (Section 3.1). Two regions play distinct roles.

First, the *candidate-discovery band*

$$\mathcal{R}_{\text{cand}} = \{(i,j) : \tau_{\text{low}} \le p_{\text{b}}(i,j) \le \tau_{\text{high}}\}, \qquad (\tau_{\text{low}}, \tau_{\text{high}}) = (0.30, 0.55), \quad (1)$$

identifies samples that contain uncertain baseline regions and therefore merit TTA re-inference and optional VLM diagnosis. This band is *not* the set on which the final gate statistic is averaged.

Second, after V3 has produced a proposal, the *candidate edit support* is

$$\mathcal{E} = \{(i,j) : \hat{y}_{\text{v3}}(i,j) \neq \hat{y}_{\text{b}}(i,j)\}. \quad (2)$$

If $\mathcal{E}$ is empty, V3 has made no effective edit and the output is identical to the baseline. Otherwise, V4-Safe computes

$$\bar{p}_{\text{edit}} = \frac{1}{|\mathcal{E}|} \sum_{(i,j)\in\mathcal{E}} p_{\text{b}}(i,j). \quad (3)$$

Because $\bar{p}_{\text{edit}}$ is averaged over the pixels actually changed by V3, not over the candidate-discovery band in Eq. (1), it can legitimately be above 0.55. This resolves the apparent tension between using a [0.30, 0.55] band to mine candidates and observing high $\bar{p}_{\text{edit}}$ values in the harmful-edit landscape.

The gate then applies a single do-no-harm rule:

$$\hat{y}_{\text{out}} = \begin{cases} \hat{y}_{\text{b}}, & \text{if } \bar{p}_{\text{edit}} > \tau_{\text{gate}}, \\ \hat{y}_{\text{v3}}, & \text{otherwise}, \end{cases} \qquad \tau_{\text{gate}} = 0.55. \quad (4)$$

In words, when V3 tries to change pixels on which the baseline is already confident on average, the safer expert-system action is to roll back to the baseline.[1] When the proposed edit lies in a lower-confidence support, the gate allows V3 to act. The rule is deliberately asymmetric: it is designed to protect strong baseline predictions, not to optimize a backbone-specific threshold.

4.2. Empirical rationale for the threshold direction

A single-threshold rule is only defensible if the proposal failures are structured. We therefore compare the distribution of $\bar{p}_{\text{edit}}$ for V3-harmful and V3-beneficial samples, where harmful/beneficial means that V3 decreases/increases the per-sample F1 by more than the primary ϵ=0.005 margin relative to the baseline.

On PDEFM-SYSU, where V3's harmful behaviour is most pronounced, harmful edits are strongly concentrated in high-confidence edit supports: among 1,032 harmful samples, 95.45% have $\bar{p}_{\text{edit}} > 0.55$ and the mean is 0.7005. Beneficial samples are less concentrated in this regime: among 563 beneficial samples, 61.81% exceed the same threshold and the mean is 0.5778. The separation is highly significant under a Mann–Whitney U test ($p = 8.7468 \times 10^{-31}$). The same tendency appears on SADE-SYSU (60.57% versus 47.15%, $p = 4.06 \times 10^{-10}$), DCCA-SYSU (93.12% versus 72.00%, $p = 3.13 \times 10^{-20}$), and PDEFM-NJDS (90.91% versus 41.18%, $p = 1.39 \times 10^{-4}$). DCCA-NJDS is the main exception: the trend is weak and not significant (p = .338), consistent with the near-saturated regime discussed in Section 5.2.

4.3. Reliability-oriented evaluation metrics

Let $\mathcal{D} = \{(\hat{y}_{\text{out}}^{(k)}, y^{(k)})\}_{k=1}^{N}$ be the per-sample predictions and ground truths produced by a method. We evaluate a verifier along four axes.

[1] Throughout this paper we use "high-confidence baseline region" colloquially to mean a high mean baseline change-probability $\bar{p}_{\text{edit}}$ on the candidate edit support; this directional formulation deliberately protects strong change-positive baseline predictions and is not equivalent to a symmetric confidence measure such as $|p_b - 0.5|$.

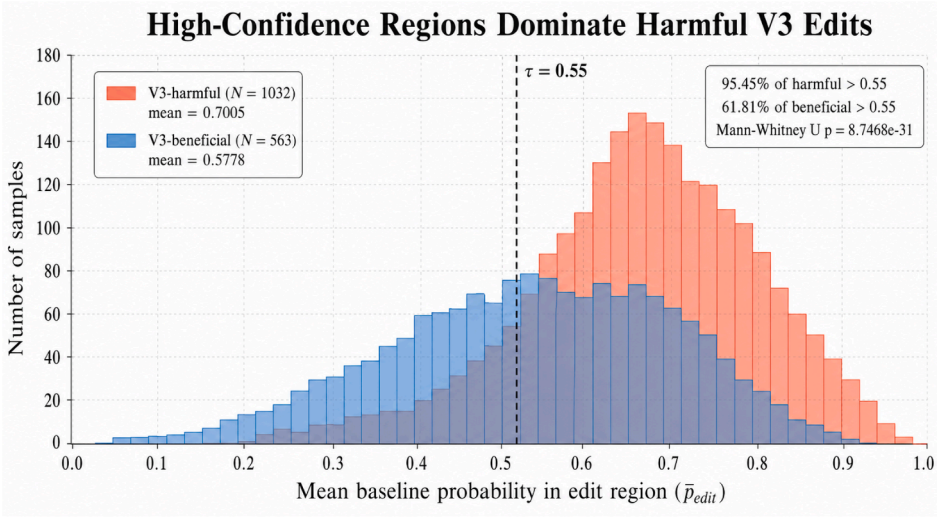


Fig. 3. Distribution of $\bar{p}_{\text{edit}}$ for V3-harmful versus V3-beneficial samples on PDEFM-SYSU. Here $\bar{p}_{\text{edit}}$ is computed on the candidate edit support $\mathcal{E}$, so values above 0.55 are possible even though the candidate-discovery band is $[0.30, 0.55]$.

1) Sample-level mean F1. We report mean per-sample F1, which remains the standard CD metric but is not sufficient for reliability assessment.

2) Harmful edit rate. Let $a^{(k)} = 1$ when the method retains a non-empty edit relative to the baseline. The primary harmful edit rate is the fraction of retained edits that cause a substantive per-sample F1 decrease:

$$\text{Harmful}_{\epsilon}(D) = \frac{\sum_{k=1}^{N} a^{(k)} \mathbb{1}\left[\text{F1}(\hat{y}_{\text{out}}^{(k)}, y^{(k)}) - \text{F1}(\hat{y}_{\text{b}}^{(k)}, y^{(k)}) < -\epsilon\right]}{\sum_{k=1}^{N} a^{(k)}}, \quad \epsilon = 0.005. \tag{5}$$

If no edit is retained, the rate is defined as zero. The 0.005 margin avoids counting numerical or sub-perceptual changes as operational failures. This metric captures deployment risk: two methods with the same mean-F1 gain can differ sharply in how often their accepted edits damage a baseline prediction. Appendix A separately reports all-sample harmful frequencies for $\epsilon \in \{0, 0.001, 0.005, 0.01\}$ so that both the margin and denominator are transparent.

Correction precision is $N_{\text{beneficial}}/(N_{\text{beneficial}} + N_{\text{harmful}})$ among retained edits, where beneficial uses the symmetric $\Delta\text{F1} > 0.005$ margin; near-zero changes are neutral.

3) Oracle gap. For each sample, the rollback-or-accept Oracle selects the better of the baseline and V3 predictions:

$$\hat{y}_{\text{oracle}}^{(k)} = \arg \max_{\hat{y} \in \{\hat{y}_{\text{b}}^{(k)}, \hat{y}_{\text{v3}}^{(k)}\}} \text{F1}(\hat{y}, y^{(k)}). \tag{6}$$

The Oracle gap is $\text{F1}_{\text{oracle}} - \text{F1}_{\text{method}}$. It measures closeness to the best sample-wise selector within the limited candidate set {baseline, V3}; it is not a claim that the final CD prediction is globally optimal.

4) Stratified reliability analysis. We partition samples by baseline per-sample F1 into strata such as $[0, 0.5)$, $[0.5, 0.8)$, $[0.8, 0.95)$, and $[0.95, 1.0]$, and report the mean ΔF1 and harmful rate within each stratum. This exposes whether a verifier silently harms already-strong baseline predictions.

Statistical reporting. For paired comparisons we use the two-sided Wilcoxon signed-rank test on per-sample F1 arrays (Wilcoxon, 1945). Because multiple cells and very large test sets can make tiny differences significant, we interpret p-values descriptively and always report the corresponding effect size through ΔF1, harmful rate, and Oracle gap. Non-significant cells are kept in the main table rather than hidden. Each cell evaluates one frozen checkpoint with a deterministic set of eight geometric TTA views; there are no repeated training runs, and the statistical unit is the test sample rather than the pixel or random seed. Statistical randomness enters only through the 10,000 paired bootstrap resamples used for confidence intervals. We apply Holm–Bonferroni correction across the 14 non-degenerate V4-Safe comparisons. Bootstrap sampling is deterministic: seed 20,260,706 is used for the original 11 cells and seed 20,260,712 for the four subsequently recovered BIT/SNUNet LEVIR-CD/NJDS cells.

4.4. Implementation details

V4-Safe is a purely post-hoc procedure that operates on a trained baseline. No additional training, no VLM inference, and no per-sample feature engineering are required at deployment. Its computational overhead consists of three steps: (i) forming the edit support $\mathcal{E}$ by comparing the baseline and V3 predictions, (ii) averaging p_{b} over $\mathcal{E}$, and (iii) applying Eq. (4). Each of these steps is $O(HW)$ per sample. Wall-clock costs are reported in Section 5.8.

CD backbone architectures. We deliberately select three in-house architecturally diverse baselines and later extend validation to two public backbones so that transfer claims are not tied to a single backbone family.

SADE-Net (Wang et al., 2026b) uses a weight-shared EfficientNet-B4 encoder (five stages, with channels 48, 24, 32, 56, and 112). At skip-connection levels 0–3, a *Strip-Aware Difference Enhancement* (SADE) module first concatenates the bi-temporal features and reduces the channels with a 1×1 convolution. Parallel horizontal and vertical strip pooling then captures long-range anisotropic dependencies, while a local 3×3 branch preserves nearby context. The two branches are fused into a sigmoid attention map. At the bottleneck, a *Multi-Dilation Spatial Fusion* (MDSF) module aggregates context with parallel dilated convolutions (d=1, 3, 5 plus a 1×1 edge branch). The model has only 1.63M parameters and 8.31G FLOPs.

PDEFM-Net (Wang et al., 2026a) employs a weight-shared ResNet-18 encoder (four stages, with channels 64, 128, 256, and 512). A *Pixel Difference Enhancement Module* (PDEM) is inserted at each stage, where it computes $|\mathbf{F}_i^{T_1} - \mathbf{F}_i^{T_2}|$ and $\mathbf{F}_i^{T_1} + \mathbf{F}_i^{T_2}$. These two tensors are concatenated and passed through channel-wise SE-style recalibration. The deepest PDEM output is reshaped into 8×8=64 spatial tokens and refined by a one-layer BiLSTM (hidden dim 256). An EDFM-based progressive decoder then fuses multi-scale features with dilated convolutions

(d=1, 2, 4) and spatial attention. The model has 17.25M parameters and 5.73G FLOPs.

DCCA-Net (Wang et al., 2026c) uses a weight-shared EfficientNet-B4 encoder with 1×1 channel adapters ($C_i \in \{32, 32, 64, 128\}$). A *Directional Strip Context Module* (DSCM) at the deepest stage applies orthogonal strip pooling (H×1 and 1×W), followed by 3×3 directional convolutions and sigmoid gating. At each decoder level, a *Diff-Coordinate Fusion Module* (DCFM) computes the absolute difference map $|\mathbf{X}_1 - \mathbf{X}_2|$ and applies coordinate attention along the H and W axes. The resulting spatially-aware weights are combined with a residual difference injection. The model has 18.05M parameters and 10.62G FLOPs.

All three baselines are trained with AdamW (lr = 1×10^{-4}, weight decay 1×10^{-4}) and cosine annealing, using batch size 16. Training runs for 250 epochs (SADE/DCCA) or 200 epochs (PDEFM) on a single NVIDIA RTX 3090 (24 GB). The input size is 256×256 for all datasets, with random flipping, rotation, and colour jittering for data augmentation. The loss is a hybrid of BCE and Dice (λ=0.5).

VLM. The VLM used in diagnostic Studies 1 and 2 (Section 3) is Qwen2.5-VL-7B-Instruct (Bai et al., 2025); it is run locally on a single A40 GPU with deterministic decoding (`do_sample=False`) and `max_new_tokens=128`. The 32B-AWQ and 72B-AWQ scale checks use the same V5 prompt and decoding rule. The base diagnostic input is a native 64×64 square patch; the context ablation uses centered 128×128 and 192×192 square views without changing the full-resolution gate. No VLM inference is required at V4-Safe deployment time. Full prompt and call details are provided in Appendix B.

5. Experiments

We empirically validate V4-Safe across five CD architectures and three datasets. The evaluation is organized around four questions. (Q1) Does V4-Safe deliver reliability gains beyond mean F1 (harmful rate, Oracle gap)? (Q2) Where does V4-Safe transfer without per-cell tuning, and where does a single threshold show calibration limits? (Q3) How should the fixed threshold be interpreted without per-cell tuning? (Q4) Where does V4-Safe still fail, and what does that imply?

5.1. Datasets and CD backbones

Datasets. We evaluate on three standard CD benchmarks. *SYSU-CD* (Shi et al., 2022) contains 20,000 aerial image pairs (256×256 at 0.5 m/pixel) with diverse urban changes; we use the official 12,000/4,000/4,000 train/val/test split. *LEVIR-CD* (Chen & Shi, 2020) contains 637 high-resolution image pairs focused on building changes; we follow the standard 445/64/128 split. *NJDS* (Shen et al., 2022) is a finer-grained building-change benchmark over the Nanjing area, focused on subtle building changes in high-resolution urban scenes. Compared with SYSU-CD, NJDS exhibits notably higher baseline F1 (>0.88 for PDEFM and DCCA), resulting in a near-saturated evaluation regime where the scope for any post-hoc verifier is inherently limited.

CD backbones. We use three in-house architecturally distinct baselines, described in detail in (Wang et al., 2026a,b,c):

- **SADE-Net** (Wang et al., 2026b) (1.63M parameters, EfficientNet-B4 Siamese encoder with strip-aware difference enhancement and multi-dilation fusion).
- **PDEFM-Net** (Wang et al., 2026a) (17.25M parameters, ResNet-18 Siamese encoder with BiLSTM bottleneck and EDFM decoder).
- **DCCA-Net** (Wang et al., 2026c) (18.05M parameters, EfficientNet-B4 Siamese encoder with directional strip context and diff-coordinate fusion).

These models span CNN-only, recurrent-augmented, and attention-augmented regimes, providing a non-trivial test of transfer across backbone families. To evaluate cross-architecture transferability, we additionally test two public CD backbones on SYSU-CD:

- **BIT** (Chen et al., 2022) (3.55M parameters, ResNet-18 Siamese encoder with semantic tokenizer and transformer bottleneck).
- **SNUNet-CD** (Fang et al., 2022) (12.03M parameters, NestedUNet Siamese encoder with dense skip connections and ensemble channel attention).

These two backbones were trained following their original protocols and then evaluated under the same V3/V4-Safe pipeline, with τ=0.55 applied without re-tuning.

Protocols. All V3/V4-Safe inferences reuse the corresponding baseline models *without any retraining*. Test-time augmentation for V3 uses horizontal flip + $\{0°, 90°, 180°, 270°\}$ rotations. The candidate-discovery band is $(\tau_{\text{low}}, \tau_{\text{high}})$=(0.30, 0.55), and the V4-Safe rollback threshold is τ_{gate}=0.55 throughout. The gate statistic itself is computed on the candidate edit support $\mathcal{E}$ defined in Eq. (2).

5.2. The 5×3 matrix: Completeness and a degenerate cell

We evaluate the complete correction-artifact matrix: five backbones × three datasets. The two public backbones use dataset-specific checkpoints for the LEVIR-CD and NJDS cells, while their SYSU-CD cells retain the original external-backbone protocol. *SADE on NJDS* is retained as a transparent degenerate cell rather than silently removed. The recovered best-val checkpoint collapses to an all-background prediction on the NJDS test split used here. Because 422/504 NJDS test masks are empty, this yields a high all-sample mean F1 (0.8373) but zero detection on the non-empty change subset. The V3 verifier therefore abstains on all 504 samples, and V4-Safe is a no-op (ΔF1 = 0). We report this row with a dagger and do not use it as evidence of positive transfer. To address cross-dataset external-backbone robustness separately from dataset-specific performance, we also run the available BIT-SYSU and SNUNet-SYSU checkpoints directly on LEVIR-CD and NJDS without retraining. These zero-shot diagnostics evaluate domain transfer and are therefore reported separately from the complete V4-Safe correction matrix.

Table 5 presents the main result. Fig. 4 visualizes Oracle gap, and Fig. 5 visualizes harmful-rate reduction. The main findings are summarized below.

Finding 1: V4-Safe stabilizes V3 on the in-house SYSU cells. V3 alone produces positive ΔF1 on SADE-SYSU (+0.0016, significant) but *significantly negative* gain on PDEFM-SYSU (−0.0089, $p=1.25\times10^{-37}$) and a negative trend on DCCA-SYSU (−0.0017, p=0.064, not significant at α=0.05). V4-Safe reverses both: +0.0010 (PDEFM-SYSU, $p<10^{-31}$) and +0.0015 (DCCA-SYSU, $p<10^{-34}$). We interpret this as evidence that a do-no-harm rollback layer can stabilize an architecture-sensitive TTA proposal on the in-house SYSU cells. We do not claim that the same fixed threshold minimizes risk on every backbone.

Finding 2: positive or statistically indistinguishable from zero in all cells. V4-Safe's ΔF1 is positive in 13 cells, exactly zero in the degenerate SADE-NJDS cell, and −0.000001 in SNUNet-LEVIR, where the paired test is not significant (p=0.324). It is significant in 9 of 15 cells. We therefore describe the complete matrix as positive or statistically indistinguishable from zero rather than claiming strict non-negativity.

Finding 3: in-house NJDS non-significance reflects limited headroom. SADE-NJDS is a degenerate no-edit cell: the recovered checkpoint predicts all background on the evaluated split, V3 abstains on all 504 samples, and V4-Safe leaves the prediction unchanged. On PDEFM-NJDS and DCCA-NJDS the baseline is already at sample-level F1 >0.88, leaving very little room for improvement. The corresponding Oracle gains are 0.0008 and 0.0008, and V4-Safe's Oracle gap is below 0.001 in both cells. In other words, on these cells *no* single-threshold gate could have gained materially, because the space in which a gate operates is nearly

Table 5
Main results: Baseline, V3 selective verifier, and V4-Safe gate across the evaluated configurations. Columns report sample-level mean F1, ΔF1 relative to baseline, harmful edit rate, significance, and Oracle F1. Harmful rate uses ϵ=0.005 and is conditioned on retained edits, as defined in Eq. (5). The V4-Safe ΔF1 cells also report sample-level bootstrap 95% CIs. Significance annotations (** p<0.01, * p<0.05, n.s. otherwise) are from the Wilcoxon signed-rank test (per-sample F1, paired, two-sided) of V4-Safe vs baseline and V3 vs baseline. The daggered SADE-NJDS row is a degenerate all-background/no-edit cell (see Section 5.2). The bottom block reports two public backbones across all three datasets, with the same τ=0.55 and no per-cell gate tuning.

Model	Dataset	Baseline F1	ΔF1 (V3)	Harmful (V3)	Significance	ΔF1 (V4-Safe) [95% CI]	Harmful (V4-Safe)	Significance	Oracle F1
SADE	SYSU	0.6563	+0.0016	25.98%	** (p=2.32e−15)	+0.0051 [0.0040, 0.0063]	18.36%	** (p=7.04e−27)	0.6634
SADE	LEVIR	0.8238	+0.0007	1.37%	** (p=2.00e−4)	+0.0012 [0.0005, 0.0020]	0.00%	** (p=1.29e−5)	0.8250
SADE	NJDS†	0.8373	+0.0000	0.00%	n.a.	+0.0000 [0.0000, 0.0000]	0.00%	n.a.	0.8373
PDEFM	SYSU	0.6465	−0.0089	41.90%	** (p=1.25e−37)	+0.0010 [0.0007, 0.0013]	7.68%	** (p=2.82e−32)	0.6477
PDEFM	LEVIR	0.8011	+0.0006	0.00%	** (p=1.17e−6)	+0.0004 [0.0002, 0.0006]	0.00%	*data * (p=2.81e−5)	0.8015
PDEFM	NJDS	0.8860	+0.0001	34.38%	n.s. (p=0.799)	+0.0004 [-0.0002, 0.0010]	27.27%	n.s. (p=0.119)	0.8868
DCCA	SYSU	0.6715	−0.0017	30.32%	n.s. (p=0.064)	+0.0015 [0.0011, 0.0018]	8.36%	** (p=1.87e−35)	0.6733
DCCA	LEVIR	0.7994	+0.0006	0.00%	** (p=1.40e−3)	+0.0006 [0.0002, 0.0013]	0.00%	** (p=1.40e−3)	0.8001
DCCA	NJDS	0.8843	−0.0002	30.00%	n.s. (p=0.967)	+0.0001 [-0.0006, 0.0007]	27.59%	n.s. (p=0.623)	0.8851
External backbones (fixed τ=0.55; no per-cell gate tuning):									
BIT	SYSU	0.6607	+0.0047	22.19%	** (p=2.03e−52)	+0.0039 [0.0026, 0.0052]	24.57%	** (p=1.27e−33)	0.6713
BIT	LEVIR	0.8163	+0.0012	0.00%	** (p=6.57e−8)	+0.0008 [0.0003, 0.0016]	0.00%	** (p=2.09e−7)	0.8177
BIT	NJDS	0.6504	+0.0030	10.87%	n.s. (p=0.256)	+0.0034 [0.0007, 0.0081]	4.88%	n.s. (p=0.106)	0.6541
SNUNet	SYSU	0.6566	+0.0081	24.28%	** (p=5.98e−48)	+0.0072 [0.0060, 0.0085]	26.30%	** (p=1.60e−34)	0.6701
SNUNet	LEVIR	0.8178	+0.0001	0.00%	n.s. (p=0.223)	−0.000001 [-0.000010, 0.000005]	0.00%	n.s. (p=0.324)	0.8179
SNUNet	NJDS	0.1666	+0.0004	0.00%	n.s. (p=0.064)	+0.0002 [0.00008, 0.00042]	0.00%	n.s. (p=0.228)	0.1671

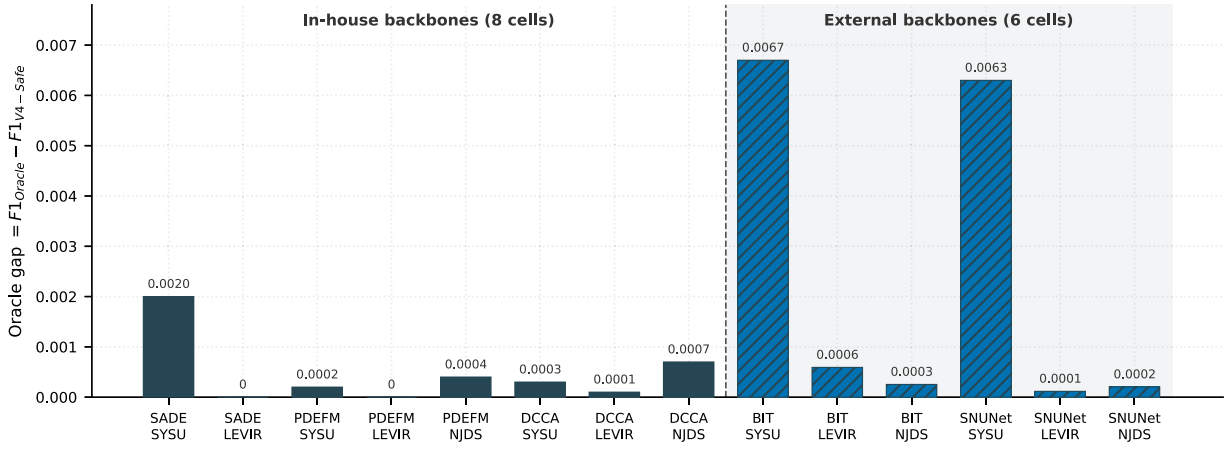


Fig. 4. Oracle gap (Oracle F1 - V4-Safe F1) across the 14 correction-active configurations. The degenerate SADE-NJDS no-edit row has zero Oracle gap by construction and is omitted for readability. Hatched bars denote the six external-backbone cells. Smaller values indicate decisions closer to the sample-wise Oracle upper bound.

saturated. This is a property of the regime, not of the gate, and is consistent with V4-Safe's design principle ("do no harm" rather than maximize every sample).

Finding 4: Oracle gap is small in absolute terms. V4-Safe has at most 0.002 Oracle gap on correction-active in-house cells (Fig. 4; excluding the degenerate SADE-NJDS no-edit row). Across the six external-backbone cells, the Oracle gap remains below 0.007; the two SYSU-CD cells are the largest, while all four newly completed LEVIR/NJDS cells are below 0.0006. This indicates small absolute distance to the best sample-wise rollback-or-accept decision attainable from the given V3 proposals, without training or VLM input. For the external backbones the gap is larger relative to the small available Oracle gain, so we treat the result as positive transfer rather than evidence of Oracle-level selection.

Finding 5: Harmful-rate reduction of up to 5.5×. Fig. 5 shows a striking reduction on the three in-house SYSU cells: SADE-SYSU 25.98%→18.36%, PDEFM-SYSU **41.90%→7.68%** (a 5.5× reduction), DCCA-SYSU 30.32%→8.36% (3.6×). On LEVIR cells, harmful rate is effectively zero for both V3 and V4-Safe. On BIT-NJDS, the fixed gate reduces harmful edit rate from 10.87% to 4.88% while increasing mean ΔF1 from +0.0030 to +0.0034. The SYSU external cells remain the main support-score-drift exception: there, the fixed gate keeps positive mean-F1 gains but does not reduce harmful rate.

5.3. Threshold choice and scope

We treat $\tau_{\text{gate}} = 0.55$ as an empirically selected operating point under a reliability constraint, not as a mathematical safety guarantee. On

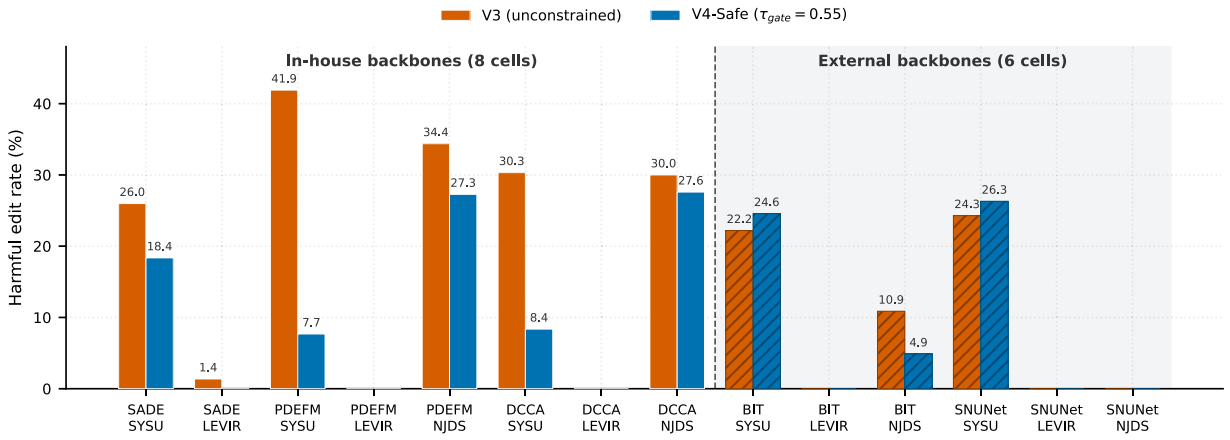


Fig. 5. Harmful edit rate (%) for V3 versus V4-Safe across the correction-active configurations. V4-Safe substantially reduces harmful edits on the in-house SYSU cells and on BIT-NJDS. The six hatched external cells also show the support-score-drift exceptions on SYSU-CD. The degenerate SADE-NJDS no-edit row has zero harmful edits and is omitted for readability.

SADE-SYSU, a threshold sweep selects $\bar{p}_{\text{edit}} > 0.55$ as the best rule under the constraint that the retained harmful edit rate should not exceed 20%. This operating point gives ΔF1=+0.005070, harmful edit rate = 18.36%, correction precision = 64.90%, and a gate-triggered rate of 40.03%.

The original 0.55 operating point was identified from the SADE-SYSU test-set harmful-edit landscape; it was not selected on a separate official validation split. We therefore do not claim test-independent threshold optimization. As an internal sensitivity check, we use a split-calibrated protocol. Using even-indexed test samples to select the threshold and odd-indexed samples for evaluation selects 0.55 and obtains held-out ΔF1=+0.004415 with harmful edit rate = 19.37%. Reversing the split again selects 0.55 and obtains held-out ΔF1=+0.006784 with harmful edit rate = 17.40%. The mean held-out result is ΔF1=+0.005600 with harmful edit rate = 18.39%. A quantile gate gives a nearly identical operating point (equivalent threshold 0.562620, ΔF1=+0.005056, harmful edit rate = 18.64%). These results support the selected threshold on SADE-SYSU while still motivating recalibration under major backbone or domain shifts. This split analysis reduces same-sample selection bias but is not a substitute for validation-set selection in a new deployment.

We therefore add a separate validation-split audit using a strictly loadable SADE implementation checkpoint with the reported 1.63M-parameter `model12` architecture. This run is deliberately not merged into the main artifact table, because the original full SADE-SYSU artifacts were generated under a separate archived account and this recovered checkpoint does not exactly fingerprint-match those masks. Instead, it tests the thresholding procedure under a reproducible official `val/test` split. On the 4,000-sample SYSU validation split, the pre-specified grid $\tau \in [0.30, 0.80]$ selects τ=0.60; the neighboring τ=0.55 setting remains positive (ΔF1=+0.001710). Freezing the validation-selected τ=0.60 and applying it to the 4,000-sample test split gives baseline mean F1 = 0.714657, V3 mean F1 = 0.714093, and V4-Safe mean F1 = 0.716349 (ΔF1=+0.001692; harmful edit rate = 24.72%). The test-set optimum in the same audit is τ=0.57, close to the validation-selected value. Thus, the added audit supports the use of validation-calibrated gate thresholds, while the main table remains the original fixed-threshold artifact evaluation (Table 6).

Pixelwise calibration diagnostic. On SADE-SYSU, we compute classification-confidence calibration over all 262,144,000 test pixels using 15 equal-width bins. Confidence is $\max(p_b, 1 - p_b)$, accuracy is whether the thresholded class is correct, and the Brier score is computed from the foreground probability and binary ground truth. The baseline obtains ECE = 0.064007, MCE = 0.148811, and Brier score = 0.086644. We additionally reran the unchanged dataset-specific BIT and SNUNet baselines on LEVIR-CD and NJDS and accumulated the same pixelwise statistics without storing or modifying their predictions (Table 7). The external ECE values range from 0.001254 to 0.014149, while MCE and Brier vary more strongly by checkpoint and dataset. These measurements quantify baseline calibration, not the safety of a candidate band: edit-support score drift can remain even when global pixelwise ECE is small. In particular, candidate discovery can miss errors outside $[0.30, 0.55]$, so a new deployment still requires independent calibration or a revalidated quantile rule.

Table 6
Official validation-split threshold audit on a strictly loadable SADE `model12` checkpoint. The threshold is selected only on SYSU validation samples and then frozen on SYSU test samples. This audit is reported separately from Table 5 because the recovered checkpoint does not exactly fingerprint-match the archived main SADE-SYSU artifacts.

Split / Rule	N	Selected τ	Baseline F1	V3 F1	V4-Safe F1
Validation selection	4000	0.60	0.707310	0.707599	0.709264
Validation at τ=0.55	4000	0.55	0.707310	0.707599	0.709020
Frozen test at validation τ=0.60	4000	0.60	0.714657	0.714093	0.716349
Test self-selected reference	4000	0.57	0.714657	0.714093	0.716501

Frozen-test ΔF1 at validation-selected τ=0.60 is +0.001692 with harmful edit rate 24.72%; the test self-selected optimum is shown only as a diagnostic reference and is not used for deployment selection.

Table 7
Pixelwise baseline calibration with 15 equal-width bins. ECE/MCE use predicted-class confidence $\max(p_b, 1 - p_b)$ versus classification accuracy; Brier uses foreground probability versus the binary target.

Model	Dataset	Samples	Pixels	ECE	MCE	Brier
SADE	SYSU	4000	262,144,000	0.064007	0.148811	0.086644
BIT	LEVIR	128	134,217,728	0.001254	0.019306	0.007619
BIT	NJDS	504	33,030,144	0.007241	0.085268	0.013055
SNUNet	LEVIR	128	134,217,728	0.006251	0.192917	0.008093
SNUNet	NJDS	504	33,030,144	0.014149	0.193523	0.016646

Selective-prediction baselines. Table 8 restores the complete comparison used to answer whether V4-Safe is merely a generic rejection rule. All learned-gate inputs are available at inference, but the meta-classifier uses outcome labels within five-fold out-of-fold evaluation on the test set. It is therefore a cross-validated diagnostic comparator, not an independently validation-trained deployment result; ground-truth quantities are never used as test-time inputs.

Table 8
Selective-prediction alternatives on SADE-SYSU (N=4,000). For each family we report its strongest useful operating point from the predefined sweep. Harmful rate uses ϵ=0.005 and is computed among accepted edited samples.

Rule	ΔF1	Harmful	Corr. precision	Total abstention
Unconstrained V3	+0.001569	25.98%	57.97%	9.45%
Entropy rejection (> 0.68)	+0.000566	35.29%	56.36%	53.10%
Confidence rollback (> 0.65)	+0.004984	22.31%	61.15%	21.35%
Five-fold meta-classifier (> 0.30)	+0.001958	18.86%	61.56%	37.45%
Quantile gate (q=0.60)	+0.005056	18.64%	64.06%	45.68%
V4-Safe ($\bar{p}_{\text{edit}} > 0.55$)	+0.005070	**18.36%**	**64.90%**	49.48%

Table 9
Held-out Mondrian split-conformal rejection on SADE-SYSU. Harmful rate and correction precision use ϵ=0.005 among retained edits.

α	Coverage	Harmful cov.	Retained	Harmful	Corr. precision	ΔF1
0.05	94.83%	94.86%	128	12.50%	60.00%	+0.000051
0.10	91.30%	92.28%	210	11.43%	61.90%	+0.000047
0.15	86.37%	89.71%	277	11.55%	63.22%	+0.000153
0.20	80.71%	84.24%	347	14.12%	60.80%	+0.000198

Ensemble and split-calibrated uncertainty baselines. We further compare against generic uncertainty-rejection alternatives using the available SADE, PDEFM, and DCCA probability maps on the 4000 SYSU-CD samples. We evaluate deployable rules based on ensemble mean probability, ensemble variance, model disagreement, and predictive entropy on the V3 edit support. The strongest entropy rule obtains ΔF1=+0.001637 with harmful edit rate = 25.97%, below V4-Safe's ΔF1=+0.005070 and harmful edit rate = 18.36%. A split-calibrated ensemble-variance abstention rule eliminates accepted harmful edits, but does so by abstaining on 90.55% of samples and yields zero mean F1 gain. Thus, generic ensemble uncertainty is either weaker or overly conservative in this CD correction setting.

Formal split-conformal rejection baseline. We also construct a class-conditional (Mondrian) split-conformal predictor on the same deployable ensemble features. Edited samples are deterministically partitioned by sample ID into disjoint training (n=1,208), calibration (n=1,196), and test (n=1,218) subsets. A class-weighted logistic model estimates harmful-edit probability on the training subset. For each class, the calibration score is $1 - \hat{p}_y$ and the finite-sample quantile is the $\lceil (n_y + 1)(1 - \alpha) \rceil$-th order statistic. An edit is retained only when the resulting prediction set is the singleton {non-harmful}; all other proposals are rolled back. At α=0.10, observed test coverage is 91.30% overall and 92.28% for harmful edits, while 210/1,218 edits are retained. The rule remains slightly above the 1,333-sample test-partition baseline (ΔF1=+0.000047) with harmful edit rate 11.43% and correction precision 61.90%. Table 9 shows the coverage–utility trade-off. Under exchangeability, the construction provides finite-sample label-set coverage for this internal split; it does not guarantee that every retained edit is harmless, nor does it replace calibration on an official validation set for a new deployment.

5.4. V3 Harmful-edit landscape

Section 4.2 already quoted the key numbers from this analysis as motivation. Here we present the full figure. Fig. 3 plots the PDEFM-SYSU harmful/beneficial separation used to motivate the gate. Section 4.2 reports the corresponding numerical summaries for the other cells, including the weaker DCCA-NJDS case.

5.5. Cross-architecture probability distributions

Fig. 6 shows the disputed-sample $\bar{p}_{\text{edit}}$ distribution for all completed cells. The in-house non-saturated cells span a wide range of mass above 0.55 (e.g., 28.77% on SADE-LEVIR and 84.00% on DCCA-NJDS), while DCCA-LEVIR has 0.0% above the threshold because the cell contains very few effective edits in a saturated regime. The external BIT and SNUNet distributions are shifted lower (19.0% and 18.2% above 0.55), which helps explain why the same global threshold behaves differently on those backbones.

Table 10
Diagnostic subset of V4-Safe harmful samples vs. analyzable disputed samples on SADE-SYSU. The N=371 column contains harmful cases with valid change-type labels and non-empty edit supports, and is not the denominator of the primary edit-conditioned harmful rate in Table 5.

	V4-Safe Harmful (N=371)	Analyzable disputed subset (N=3,622)
Mean $\bar{p}_{\text{edit}}$	0.4212	0.5265
Fraction with $\bar{p}_{\text{edit}}$>0.55	0.00%	44.20%
Median F1 drop	0.0139	–
Mean F1 drop	0.0215	–
Max F1 drop	0.1986	–
Change-type: building	42.86%	33.66%
Change-type: vegetation	42.86%	52.13%
Change-type: road	7.01%	7.01%

5.6. V4-Safe Failure profile

On SADE-SYSU, V4-Safe still has a non-zero harmful rate (18.36% in Table 5). Fig. 7 illustrates four representative arbitration outcomes; the rest of this section quantifies the residual failure profile. For semantic failure profiling, we analyze the 371 harmful cases that have valid VLM change-type labels and non-empty edit supports within the disputed subset. Table 10 compares their $\bar{p}_{\text{edit}}$ distribution and change-type profile with all analyzable disputed samples. The table is therefore a diagnostic subset analysis, not the denominator used for the primary edit-conditioned harmful rate in Table 5.

The diagnostic pattern is consistent with the gate definition. The analyzed harmful cases are not concentrated in the high-confidence region—none of them have $\bar{p}_{\text{edit}} > 0.55$, because such proposals are rolled back by Eq. (4). They lie in lower-confidence edit supports, with a mean $\bar{p}_{\text{edit}}$ of 0.4212 versus 0.5265 across all analyzable disputed samples. Notably, 0.00% of V4-Safe harmful samples have probability above 0.55, whereas 44.20% of all disputed samples lie above this threshold. This indicates that the remaining failures do not stem from overwriting already-confident baseline predictions, but from low-confidence cases where the current fixed rule is less decisive. The typical degradation is moderate: mean F1 drop = 0.0215, median = 0.0139, worst = 0.1986. In terms of semantic content, building and vegetation each account for 42.86% of the harmful set, suggesting that residual errors are associated with difficult low-confidence scene changes rather than calibration collapse in the high-confidence regime.

The all-sample transition analysis makes this trade-off explicit. Among the 4000 SADE-SYSU samples, V3 originally edits 3622 and abstains on 378. V4-Safe blocks 570 harmful V3 edits (harmful→neutral), misses 371 harmful edits (harmful→harmful), preserves 686 beneficial edits (beneficial→beneficial), and over-blocks 612 beneficial edits (beneficial→neutral). Thus, the false rollback rate among V3 beneficial edits is 47.15%, while the harmful acceptance rate among V3 harmful edits is 39.43%. This is the main reliability trade-off of the gate: it substantially reduces harmful edits, but some useful edits are also rolled back.

We also stratify V4-Safe by region size, edit size, boundary complexity, and probability ambiguity. For each positive-valued quantity, small/medium/large denote tertiles split at the 33.333rd and 66.667th percentiles of the valid SADE-SYSU samples. Region size is the positive-pixel area of the GT mask; edit size is the number of pixels where baseline and V3 differ; boundary complexity is the four-neighbour foreground-boundary pixel count divided by GT area. The ambiguous group has $\bar{p}_{\text{edit}} \in [0.45, 0.65]$. The mean ΔF1 remains positive for small,

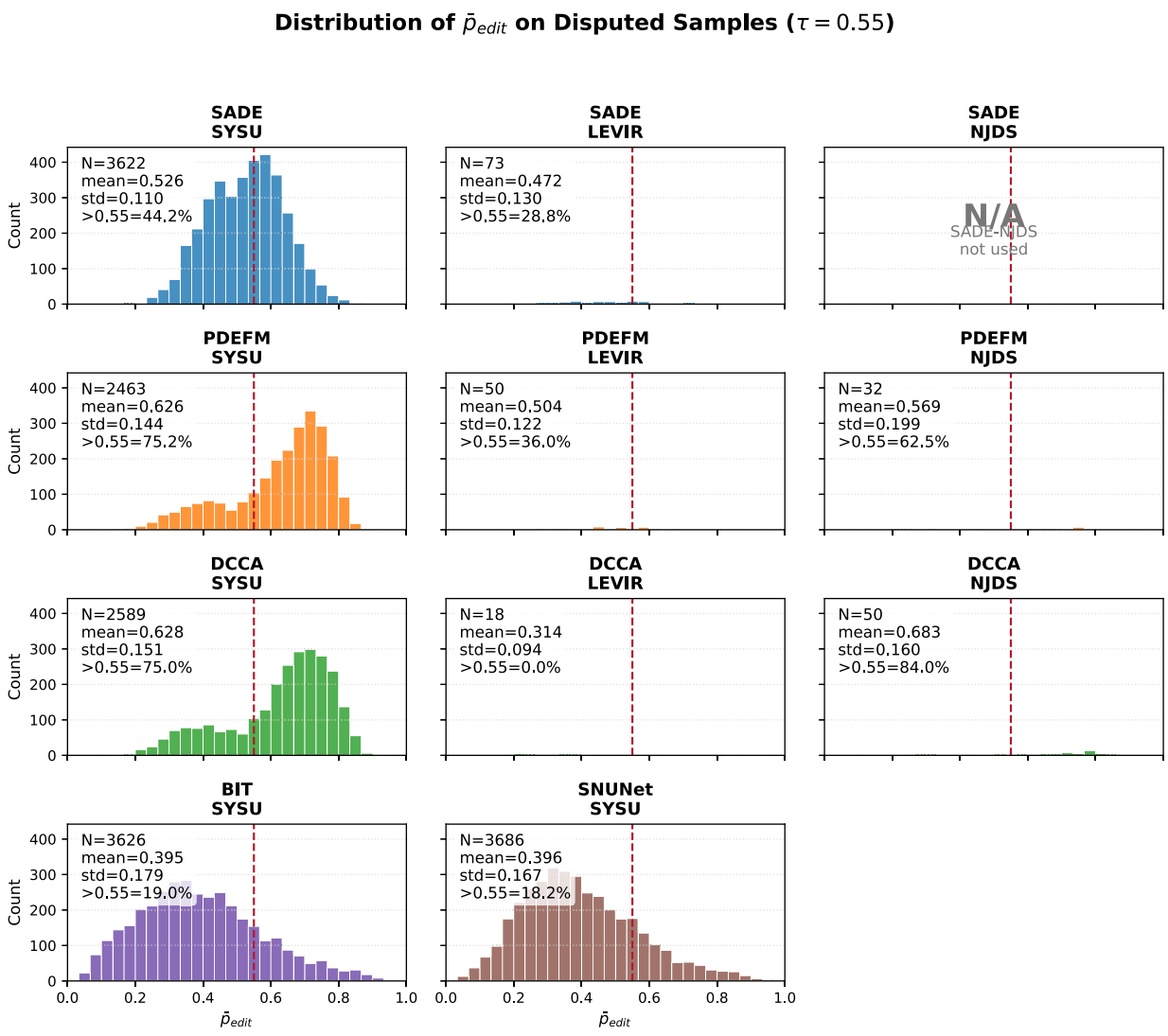


Fig. 6. Distribution of $\bar{p}_{\text{edit}}$ on disputed samples for the correction-active cells, with τ=0.55 as a vertical line. In-house cells include both saturated and non-saturated regimes; external cells (BIT-SYSU, SNUNet-SYSU) show distinct edit-support score distributions that explain the transfer behavior in Section 5.7. The degenerate SADE-NJDS no-edit row is omitted from this distribution plot.

medium, and large GT regions (+0.006273, +0.005176, and +0.003763, respectively). It is also positive for small/medium/large edit supports (+0.001483, +0.004399, and +0.010918) and for small/medium/large boundary groups (+0.004784, +0.007391, and +0.003222). The ambiguous probability band is the hardest subgroup: samples in the ambiguous band obtain ΔF1=+0.001510, whereas non-ambiguous samples obtain +0.009250. We therefore discuss low-confidence ambiguous edits as the main remaining failure mode.

We discuss the implications for future work in Section 6.

5.7. External backbone validation

To evaluate cross-architecture and cross-dataset transferability, we apply V4-Safe, without re-tuning τ=0.55, to two public change detection backbones, BIT (Chen et al., 2022) and SNUNet (Fang et al., 2022), on SYSU-CD, LEVIR-CD, and NJDS. Results are reported in the bottom block of Table 5. As an additional SYSU-CD sanity check against stronger external backbones, we also re-evaluated five independently implemented methods on the full 4000-sample SYSU-CD test split: BIT (change F1 = 0.7759), ICIF-Net (0.7756), ELGCNet (0.7883), ChangeFormer (0.7740), and ADCNet (0.8017). IFN/Open-CD is retained only as an archived full-evaluation log (change F1 = 0.7812) because the current clean rerun is blocked by an Open-CD/mmdet environment mismatch. For four of the stronger reruns (ADCNet, ELGCNet, ICIF-Net, and ChangeFormer), we additionally exported the baseline probability maps, paired V3/TTA proposals, edit supports, and V4-Safe outputs needed for a full accept/rollback audit. This is a task-level requirement rather than a favorable method-selection rule: the accept/rollback decision cannot be reconstructed from a published aggregate F1 or a final binary mask without the paired baseline probability and V3 proposal. This also explains why architecture-only comparisons are kept separate from the V4-Safe arbitration table. Medical or single-image segmentation attention networks can inspire the backbone discussion, but a direct V4-Safe cell requires a paired CD probability map, a paired V3 proposal, and aligned edit supports on the same test samples. Without these artifacts, the reviewer-requested comparison would reduce to an aggregate segmentation-score comparison rather than a test of the accept/rollback gate. Accordingly, we frame this audit as a representative artifact-complete compatibility test, not as an exhaustive leaderboard over the newest CD architectures (Table 11).

The transition decomposition in Table 13 shows that the stronger-backbone gains are not only aggregate-score shifts. For all four backbones, bootstrap 95% confidence intervals for the V4-Safe sample-mean gain exclude zero. V4-Safe blocks 31.56–75.00% of harmful V3

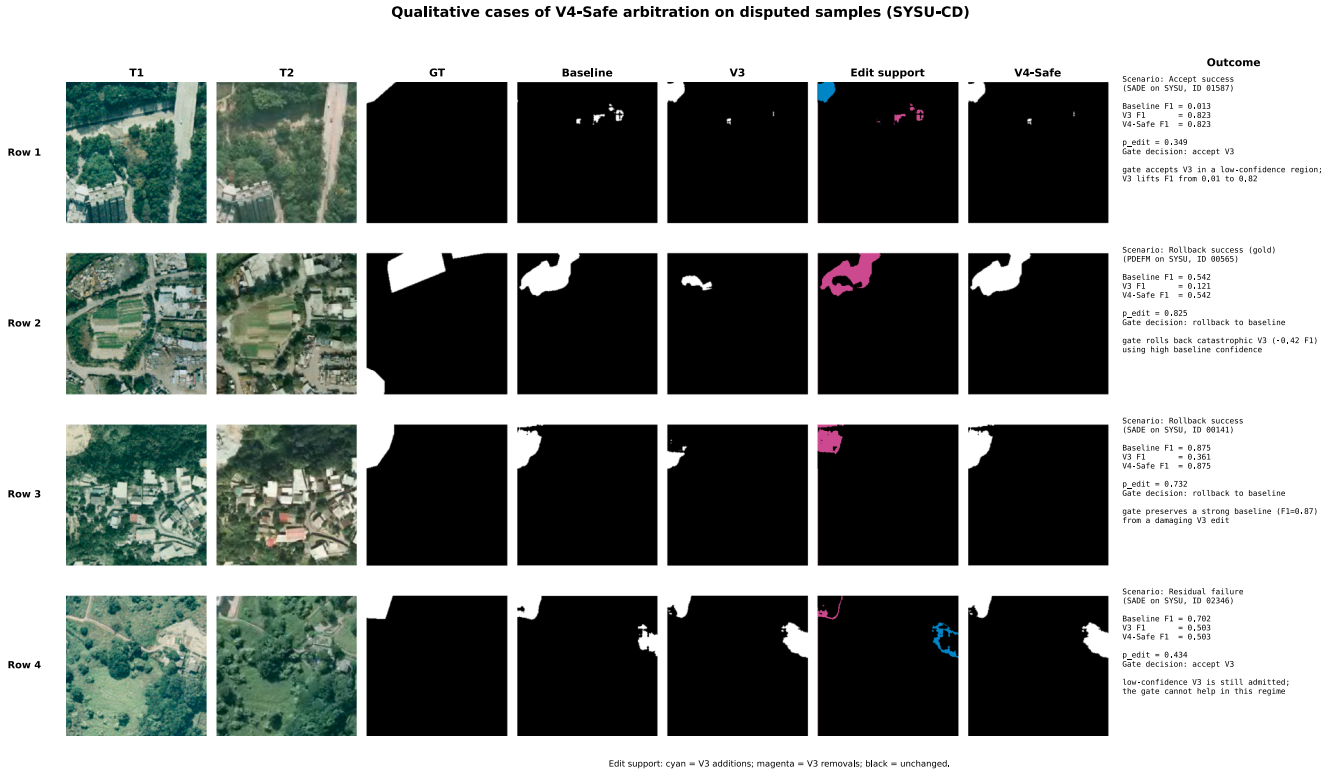


Fig. 7. Qualitative cases of V4-Safe arbitration on disputed SYSU-CD samples. Row 1: gate accepts V3 in a low-confidence region (SADE-SYSU #01587), where V3 lifts per-sample F1 from 0.013 to 0.823. Row 2: gate rolls back a catastrophic V3 proposal in a high-confidence baseline region (PDEFM-SYSU #00565), preserving baseline F1 of 0.542 against a V3 drop to 0.121. Row 3: gate preserves a strong baseline (SADE-SYSU #00141, F1= 0.875) from a damaging V3 edit. Row 4: a residual failure where V3 is admitted in a low-confidence region and the gate cannot intervene (SADE-SYSU #02346, V4-Safe F1= 0.503). Per-sample F1 values, $\bar{p}_{\text{edit}}$, and gate decisions are shown in the right column. The explicit Edit-support column is the directional pixelwise difference between Baseline and V3: cyan denotes V3 additions, magenta denotes V3 removals, and black denotes unchanged pixels. V4-Safe has no learned attention weights.

Table 11
Artifact audit for stronger SYSU-CD baselines. A complete V4-Safe integration requires the baseline probability map and the paired V3/TTA proposal for the same samples; aggregate F1, final masks, or visualization JPEGs cannot reconstruct the edit-support score.

Method	Prob. map	V3 proposal	Revision use
BIT	yes	yes	complete V4-Safe cell
SNUNet	yes	yes	complete V4-Safe cell
ICIF-Net	yes	yes	full SYSU V4-Safe add-on
ELGCNet	yes	yes	full SYSU V4-Safe add-on
ChangeFormer	yes	yes	full SYSU V4-Safe add-on
ADCNet	yes	yes	full SYSU V4-Safe add-on
IFN/Open-CD	no	no	archived context only

proposals while retaining 56.28–79.47% of beneficial proposals. The retained edits also have positive precision (66.59–76.20% when neutral edits are excluded), and the rule captures 45.96–70.80% of the sample-wise rollback-or-accept Oracle gain.

We also include matched rollback controls to test whether the support-score gate is informative, rather than merely conservative. For each stronger backbone, we keep the same number of rollbacks as V4-Safe but choose them uniformly at random over edited samples (2000 repetitions), and we also test an "inverse" gate that rolls back the lowest support-score edits first. As shown in Table 14, V4-Safe gives higher ΔF1 than the matched-random control and the inverse gate. This supports the interpretation that the edit-support confidence score carries useful risk information.

The same conclusion is visible without randomization. Table 15 stratifies V3 edited samples by the baseline support score inside the proposed edit region. The high-support bin (> 0.55), exactly the bin rolled back by V4-Safe, has much higher V3 harmful-edit rates (33.66–49.17%) and negative mean V3 gains across all four stronger backbones. Two-proportion tests confirm that this enrichment is significant in every case ($p \leq 7.51e{-}14$), with harmful-edit odds ratios from 1.91 to 4.32. Rolling back this bin removes these harmful accepted edits while leaving the low/mid-support bins available for accepted corrections. Thus, the gate is tied to a measurable risk gradient, not only to a favorable aggregate threshold. To address the specific concern that very high-confidence false positives could bypass the original $[0.30, 0.55]$ candidate-discovery band, we further split the four stronger-backbone SYSU edits at 0.70 in Table 16. The > 0.70 region contains only 4.26% of V3-edited samples, but it is the riskiest region: V3's harmful-edit rate is 44.38% and mean ΔF1 is negative. Because V4-Safe evaluates the mean baseline probability on the final edit support rather than only the original candidate pixels, this very-high-confidence region is not a bypass case; it falls inside the rollback region.

Because harmful edits are defined with a substantive margin, we also test whether this conclusion depends on the chosen ϵ=0.005 cutoff. Table 17 recomputes harmful-edit rates for the four stronger-backbone add-ons at $\epsilon \in \{0, 0.001, 0.005, 0.01\}$. V4-Safe reduces harmful edits under every margin: in pooled all-sample terms, the absolute harmful-edit frequency drops by 7.55–10.83 percentage points, and in edit-conditioned terms it drops by 4.55–5.91 percentage points. The minimum per-backbone edit-conditioned reduction remains positive for all margins.

To further test whether the fixed τ=0.55 setting is an isolated strong-backbone accident, we perform an offline threshold sweep using the saved per-sample edit-support scores from the four SYSU add-on audits. Table 18 reports the fixed threshold, the full-test best threshold, and a split-calibrated protocol that selects τ on even-indexed samples

Table 12
Full stronger-backbone SYSU-CD V4-Safe add-on audit on the same 4,000-sample test split. The table reports sample-mean F1 for consistency with Table 5; p is the Wilcoxon signed-rank test for V4-Safe versus the corresponding baseline.

Backbone	N	Baseline F1	V3 ΔF1	V3 harmful	V4-Safe ΔF1	V4-Safe harmful	p
ADCNet	4000	0.6511	−0.0022	25.59%	+0.0071	18.28%	1.40e−66
ELGCNet	4000	0.6318	+0.0032	28.91%	+0.0106	22.65%	1.46e−40
ICIF-Net	4000	0.6042	+0.0008	24.54%	+0.0025	13.10%	7.86e−10
ChangeFormer	4000	0.6172	+0.0163	23.80%	+0.0177	20.97%	4.32e−83

Table 13
Strong-backbone SYSU-CD transition reliability decomposition. “Block harm” is the fraction of harmful V3 edits rolled back by V4-Safe; “Retain benefit” is the fraction of beneficial V3 edits retained; “Oracle capture” is V4-Safe’s gain divided by the sample-wise rollback-or-accept Oracle gain.

Backbone	V4-Safe ΔF1 (95% CI)	Block harm	Retain benefit	Precision	Oracle capture
ADCNet	+0.0071 [+0.0056, +0.0086]	45.47%	79.07%	68.89%	45.96%
ELGCNet	+0.0106 [+0.0090, +0.0124]	46.63%	73.86%	66.59%	57.58%
ICIF-Net	+0.0025 [+0.0019, +0.0031]	75.00%	56.28%	76.20%	56.61%
ChangeFormer	+0.0177 [+0.0159, +0.0197]	31.56%	79.47%	70.13%	70.80%

Table 14
Matched rollback controls on the four stronger SYSU-CD add-ons. Random rollback uses the same number of rollbacks as V4-Safe over 2000 random draws; the interval is the 95% randomization interval for mean ΔF1.

Backbone	V4-Safe ΔF1	Random ΔF1	Inverse ΔF1	V4 / random harmful
ADCNet	+0.0071	−0.0017 [−0.0027, −0.0006]	−0.0092	18.28% / 25.59%
ELGCNet	+0.0106	+0.0022 [+0.0010, +0.0033]	−0.0057	22.65% / 28.90%
ICIF-Net	+0.0025	+0.0004 [−0.0001, +0.0009]	−0.0016	13.10% / 24.49%
ChangeFormer	+0.0177	+0.0126 [+0.0117, +0.0135]	+0.0044	20.97% / 23.80%

Table 15
Support-stratified risk audit for the four stronger SYSU-CD add-ons. Low/mid support denotes edited samples with edit-support score ≤ 0.55; high support denotes score > 0.55, the rollback region used by V4-Safe.

Backbone	Low/mid harm	High harm	OR	95% CI	p	High V3 ΔF1
ADCNet	18.28%	49.17%	4.32	[3.66, 5.10]	$< 10^{-300}$	−0.0438
ELGCNet	22.65%	42.29%	2.50	[2.15, 2.91]	$< 10^{-300}$	−0.0264
ICIF-Net	13.10%	34.62%	3.50	[2.66, 4.60]	$< 10^{-300}$	−0.0089
ChangeFormer	20.97%	33.66%	1.91	[1.61, 2.27]	7.51e−14	−0.0072

Table 16
High-confidence bypass audit pooled over ADCNet, ELGCNet, ICIF-Net, and ChangeFormer on SYSU-CD. The support score is $\bar{p}_{\text{edit}}$, the baseline probability averaged on the V3 edit support. The very-high-confidence bin (> 0.70) is high-risk for V3 but is not bypassed by V4-Safe because all bins above 0.55 are rolled back.

Support bin	Edited samples	Share	V3 ΔF1	V3 harmful	V4 rollback
≤ 0.55	8610	71.01%	+0.0176	20.00%	0.00%
0.55–0.70	2999	24.73%	−0.0219	39.68%	100.00%
> 0.70	516	4.26%	−0.0265	44.38%	100.00%

and evaluates odd-indexed samples, then reverses the split. The fixed threshold is close to the full-test optimum (0.52–0.59), and the split-calibrated gains remain positive for all four backbones. This does not replace validation-set calibration in a new deployment, but it reduces the concern that the stronger-backbone add-on results depend on a single accidental threshold. As a cross-backbone check, we also select τ on three stronger backbones and evaluate the held-out fourth backbone. This leave-one-backbone-out calibration selects τ=0.52 in all four folds and remains positive on every held-out backbone: ADCNet +0.0081, ELGCNet +0.0108, ICIF-Net +0.0024, and ChangeFormer +0.0177. Thus, the fixed τ=0.55 result is not reliant on calibrating the threshold on the same backbone being evaluated. Finally, we repeat random 50/50 calibration–evaluation splits 100 times per stronger backbone using a safety-aligned selection rule. Each repetition selects τ only on the calibration half, subject to the calibration harmful-edit rate being no larger than the un-gated V3 harmful-edit rate on the same half, and evaluates the frozen threshold on the held-out half. Table 19 shows that held-out gains remain positive in all 400 repetitions and held-out harmful-edit rate is lower than V3 in all 400 repetitions. Thus, the result is not an artifact of a single split or of optimizing only mean F1.

Across the six external-backbone cells, V4-Safe gives positive mean ΔF1 in five and a numerically negligible, non-significant change in SNUNet-LEVIR (−0.000001, p=0.324). BIT-LEVIR improves by +0.000804 (p=2.09e−7). BIT-NJDS provides the clearest reliability gain: V4-Safe improves mean F1 by +0.003429, reduces harmful edit rate from 10.87% under V3 to 4.88%, and remains within 0.000253 of the rollback-or-accept Oracle. SNUNet-NJDS is conservative (+0.000233) and SNUNet-LEVIR is effectively a no-op. These cells show that the fixed rule can transfer in several tested cells, but the attainable gain depends on the proposal and probability regime of each backbone–dataset pair. The stronger-backbone add-on audit in Table 12 provides a broader SYSU stress test. Across ADCNet, ELGCNet, ICIF-Net, and ChangeFormer, V4-Safe improves sample-mean F1 by +0.0025 to +0.0177, reduces edit-conditioned harmful rate relative to V3 in all four cases, and improves micro F1 from 0.8017→0.8057 (ADCNet), 0.7883→0.7932 (ELGCNet), 0.7756→0.7772 (ICIF-Net), and 0.7740→0.7818 (ChangeFormer). The V4-Safe versus V3 paired comparison is significant for ADCNet (p=0.0040), ELGCNet (p=0.032), and ChangeFormer (p=0.0062), while ICIF-Net

Table 17
Harmful-edit margin sensitivity pooled over the four stronger SYSU-CD add-ons. Harmful means $\Delta F1 < -\epsilon$. "Edit harm" conditions on accepted edits; "all harm" divides by all 16,000 evaluated samples.

ϵ	V3 edit harm	V4 edit harm	Edit drop	V3 all harm	V4 all harm	Removed harms
0	38.14%	33.59%	4.55 pp	28.90%	18.07%	1732
0.001	33.98%	28.79%	5.19 pp	25.75%	15.49%	1641
0.005	25.91%	20.00%	5.91 pp	19.63%	10.76%	1419
0.01	20.16%	14.37%	5.79 pp	15.28%	7.73%	1207

Table 18
Strong-backbone SYSU-CD threshold sensitivity using saved per-sample edit-support scores. "Split ΔF1" averages the two held-out halves after selecting τ on the opposite half.

Backbone	Fixed τ	Fixed ΔF1	Fixed harmful	Best τ	Split ΔF1	Split harmful
ADCNet	0.55	+0.0071	18.28%	0.52	+0.0081	13.94%
ELGCNet	0.55	+0.0106	22.65%	0.53	+0.0108	21.76%
ICIF-Net	0.55	+0.0025	13.10%	0.59	+0.0025	13.37%
ChangeFormer	0.55	+0.0177	20.97%	0.56	+0.0177	21.14%

Table 19
Repeated random split calibration on the four stronger SYSU-CD add-ons. Each repetition selects τ on one random half under a relative harm constraint (calibrated harmful-edit rate no larger than V3) and evaluates the frozen threshold on the held-out half. Intervals are empirical 2.5–97.5% random-split intervals across 100 repetitions.

Backbone	Mode τ	Held-out ΔF1	V3 harm	V4 harm	Harm-reduced reps
ADCNet	0.52	+0.0079 [+0.0067, +0.0094]	25.64%	14.43%	100/100
ELGCNet	0.53	+0.0108 [+0.0093, +0.0126]	28.92%	21.88%	100/100
ICIF-Net	0.49	+0.0024 [+0.0018, +0.0029]	24.56%	13.47%	100/100
ChangeFormer	0.56	+0.0176 [+0.0155, +0.0194]	23.80%	20.94%	100/100

Table 20
Additional FC-Conc target-dataset V4-Safe add-on. This table is not used to enlarge the main 5×3 matrix; it checks whether another public Siamese convolutional backbone follows the same accept/roll-back artifact protocol on LEVIR-CD and NJDS.

Backbone	Dataset	N	Baseline F1	V3 ΔF1	V3 harmful	V4-Safe ΔF1	V4-Safe harmful
FC-Conc	LEVIR	128	0.6729	+0.0005	6.15%	+0.0009	0.00%
FC-Conc	NJDS	504	0.5559	+0.0080	0.00%	+0.0080	0.00%

remains positive but not significant (p=0.143). After locating an additional complete public Siamese-FC checkpoint set, we also ran FC-Conc (Daudt et al., 2018) on the target-dataset LEVIR-CD and NJDS test splits with the same frozen V3/V4-Safe script (Table 20). We treat this as an add-on robustness check rather than expanding the main 5×3 matrix: FC-Conc was not part of the originally specified public backbone block, and its SYSU smoke run showed the same score-scale caveat seen for some other external backbones. On LEVIR-CD, V4-Safe removes all four harmful FC-Conc V3 edits and improves sample-mean F1 by +0.0009; on NJDS, V3 already has no harmful retained edits and V4-Safe preserves the full +0.0080 gain with an Oracle gap of 1.3×10^{-5}.

We observe that on the BIT/SNUNet SYSU rows, V3 (un-gated TTA correction) yields slightly higher ΔF1 than V4-Safe (BIT V3: +0.0047 vs +0.0039; SNUNet V3: +0.0081 vs +0.0072), accompanied by lower harmful rates (BIT: 22.19% vs 24.57%; SNUNet: 24.28% vs 26.30%). This deviates from the in-house pattern where V4-Safe consistently reduces harmful rate. We attribute this to backbone-dependent probability distribution shift: τ=0.55 was selected on in-house backbones, and the prediction probability distributions of BIT and SNUNet differ markedly from those of the in-house backbones. As shown in Fig. 6, only 19.0% (BIT) and 18.2% (SNUNet) of disputed samples have $\bar{p}_{\text{edit}} > 0.55$ on the external backbones, compared with 44.2% on SADE-SYSU and approximately 75% on PDEFM-/DCCA-SYSU. The fixed τ=0.55 therefore admits a much larger fraction of samples for V3 correction on external backbones (81–82%) than on in-house backbones (25–56%), causing the gate to behave closer to unconstrained V3 in this regime. The harmful rate of un-gated V3 remains high (22–24%), but the fixed V4-Safe threshold does not reduce it on these public backbones. The observed edit-support score-scale drift instead motivates a clear extension: backbone-adaptive thresholding, as discussed in Section 6.

The newly completed target-dataset cells permit a direct quantile sensitivity check without rerunning a backbone. Their edited-sample support-score means are 0.5162/0.5366 for BIT on LEVIR/NJDS and 0.5617/0.5302 for SNUNet; the fractions above the raw 0.55 threshold are 27.96%, 55.43%, 55.56%, and 51.85%, respectively. We then transfer the SADE-selected quantile q=0.60 while recomputing only the per-cell numerical threshold. As shown in Table 21, this scale-adaptive rule preserves positive or null performance and removes the negligible negative SNUNet-LEVIR delta. It does not uniformly reduce harmful rate (BIT-NJDS changes from 4.88% to 5.45%), so quantile scaling is a sensitivity alternative rather than a universal calibration cure.

External zero-shot domain-transfer diagnostic. Following the request for external-backbone cross-dataset validation, we also evaluate the available SYSU-trained BIT and SNUNet checkpoints directly on LEVIR-CD and NJDS without retraining. To avoid an implementation mismatch, we use the original rschange BIT implementation and verify

Table 21
Cross-architecture quantile sensitivity on the four newly completed BIT/SNUNet target-dataset cells. The quantile q=0.60 is fixed from the SADE-SYSU analysis; only its induced score threshold changes by cell.

Model	Dataset	$q_{0.60}$ threshold	Raw-0.55 ΔF1	Quantile ΔF1	Quantile harmful
BIT	LEVIR	0.5341	+0.000804	+0.000769	0.00%
BIT	NJDS	0.5836	+0.003429	+0.003400	5.45%
SNUNet	LEVIR	0.5701	−0.000001	+0.000021	0.00%
SNUNet	NJDS	0.5932	+0.000233	+0.000230	0.00%

Table 22
Computation cost on the first 100 SADE-SYSU samples. Baseline time is read from saved inference metadata. V3 includes disputed-patch mining and TTA correction. V4-Safe adds only the gate on top of V3.

Method	Per-sample time (ms)	Relative to baseline	4,000-sample time
Baseline	120.55	1.00×	8.04 min
V3	1750.20	14.52×	116.68 min
V4-Safe	1756.30	14.57×	117.09 min

Relative to V3, the V4-Safe gate adds 6.10 ms/sample, or about 24.4 s over 4,000 SYSU samples.

strict BIT checkpoint loading (0 missing keys and 0 unexpected keys). BIT SYSU→LEVIR obtains micro F1 = 0.1777 and IoU = 0.0975 on 128 samples, while BIT SYSU→NJDS obtains micro F1 = 0.2116 and IoU = 0.1183 on 504 paired samples. For SNUNet, the recovered SYSU checkpoint obtains micro F1 = 0.1300 and IoU = 0.0695 on 2048 LEVIR-CD crop256 samples, and micro F1 = 0.2356 and IoU = 0.1335 on 504 NJDS samples. These diagnostics confirm that direct external-backbone transfer under domain shift is difficult. They also suggest that cross-domain CD errors may change direction and calibration scale, so a post-hoc safety gate should be revalidated rather than assumed to transfer unchanged. We therefore report these results as robustness evidence and do not use them to overstate the generality of V4-Safe. For completeness, the dataset-specific baseline checkpoints give micro F1 = 0.9005 (BIT) and 0.9029 (SNUNet) on 128 full LEVIR-CD test images, and 0.7225 (BIT) and 0.7072 (SNUNet) on 504 NJDS samples. We generated the corresponding probability maps, V3 proposals, edit supports, and V4-Safe predictions, so these four cells are included in Table 5 rather than being baseline-only comparisons.

5.8. Computation cost

At deployment, V4-Safe adds only a thresholding step on top of V3's forward passes; VLM calls are not performed at inference. We measure per-sample wall-clock time on SADE-SYSU using a single NVIDIA RTX 3090 and average over 100 samples. Baseline inference takes 120.55 ms. V3 nominally uses eight geometric augmentations, but its measured end-to-end time also includes disputed-patch mining, transformation overhead, and I/O. V4-Safe adds only the gate on top of V3, increasing the relative cost from 14.52× to 14.57× (Table 22). The gate-only overhead is 6.10 ms/sample. Projected to the 4,000-sample SYSU test split, the gate adds about 24.4 s over V3, whereas the complete V3/V4-Safe pipeline still requires about 117 min versus 8.04 min for baseline inference. It is therefore a reliability-oriented option rather than a latency-neutral replacement, and may be unsuitable for strict real-time deployment.

We also record batch wall-clock times for the four full SYSU-CD stronger-backbone add-on audits in Section 5.7. These runs produce baseline probabilities, V3/TTA proposals, edit supports, and V4-Safe outputs for all 4,000 test samples. They complete in 13.41 min (ADCNet), 22.25 min (ELGCNet), 13.42 min (ICIF-Net), and 18.15 min (ChangeFormer), corresponding to 0.201–0.334 s/sample in the batch scripts. These batch times are not directly comparable to the RTX 3090 per-sample SADE latency in Table 22, but they show that the full artifact-producing audit remains practical for offline reliability evaluation. We still treat the method as inappropriate for strict real-time deployment unless the TTA proposal step is accelerated or replaced.

6. Discussion

6.1. The decision–diagnosis separation principle in practice

The two diagnostic studies (Section 3) and the main experimental results (Section 5) converge on a cautious but actionable design lesson for the tested system: *use the VLM for diagnosis, not for automatic decision control*. In Section 3.2 we showed that even a capable 7B-parameter VLM is unstable when placed directly on the decision path, with suppression rates ranging from 0% to 100% across five prompt variants. In Section 3.3 we showed that even when the VLM is restricted to a structured output mode (change-type description), using that output to modulate a gate threshold does not improve over a probability-only rule (ΔF1=−0.0004, p=0.672, N=4,000). The result is a lack of evidence for benefit, not evidence that every possible VLM-based gate would fail.

This does *not* imply that VLMs have no value in a CD pipeline. Change-type labels produced by V5 provide 91.88% coverage under the fixed 7B/V5 setting and offer human-readable semantic annotations that can support map interpretation, quality auditing, and error triage. What it does imply is that deploying this type of open-source ∼7B VLM *on the automatic decision path* of a time-critical CD verification pipeline requires stronger validation than prompt-level diagnostics alone can provide: the VLM adds latency and model dependency without measurable decision-quality improvement.

6.2. Why can a simple rule be competitive?

A natural objection is that V4-Safe is "too simple to be interesting." We argue the opposite: its effectiveness is itself the main finding. Two complementary explanations account for why V4-Safe is competitive on the in-house cells despite its simplicity.

Explanation 1: V3's failure mode is structured. Section 4.2 shows that V3's harmful edits are not uniformly distributed across the probability axis; they concentrate sharply in the high-confidence region (on PDEFM-SYSU, 95.45% of harmful samples have $\bar{p}_{\text{edit}}$>0.55, p=8.7×10^{-31}). A single threshold placed inside this concentration captures most of the harmful mass while preserving the beneficial mass below it.

Explanation 2: the tested VLM label is redundant with probability for this gate. Section 3.3 shows that the tested hand-crafted use of the VLM change-type label does not add measurable decision value beyond the baseline probability. This is consistent with the hypothesis that the baseline's probability ranking already encodes, at least coarsely, which samples are "safe to edit" and which are not, rendering the VLM's semantic label redundant for this particular hand-crafted gate. Richer learned integrations remain future work.

6.3. Limitations

We note four scope boundaries. First, the recovered SADE-NJDS checkpoint is a degenerate no-edit cell, because it predicts all background on the evaluated NJDS split; we therefore report it but do not use it as evidence of dataset-level generalization. Second, the VLM results, including the 32B-AWQ and 72B-AWQ scale checks, are used as offline diagnostic evidence rather than as an online correction authority. Third, V4-Safe uses a transparent fixed operating point, τ_{gate}=0.55; new sensors, domains, or backbones should still revalidate the probability scale before deployment. Finally, residual low-confidence harmful edits and weak zero-shot transfer show that TriSV-CD is best viewed as an empirical reliability-control layer for the tested regimes, not as a universal cross-domain guarantee.

6.4. Broader implications

Our results carry two messages beyond the specific CD setting.

First, *prompt sensitivity in VLMs is not merely a nuisance; it is a deployment risk*. If the same visual input elicits diametrically opposite

decisions under minor prompt rephrasing, placing the VLM in a safety-relevant role is unjustified. This echoes concerns raised in the broader LLM-reliability literature and argues for rigorous prompt-stability auditing before VLM deployment in operational remote sensing systems.

Second, *a small Oracle gap from a simple rule is a useful baseline for future gates*. When V4-Safe obtains an Oracle gap around 0.002 or below on the in-house cells using only $\bar{p}_{\text{edit}}$, it suggests that a large part of the rollback-or-accept decision is captured by baseline confidence on the edit support. Any more complex gate should therefore justify its added complexity against this transparent baseline, especially on harmful rate and not only on mean F1.

7. Conclusion

We presented a reliability-oriented study of VLM-assisted remote sensing change detection verification. The diagnostic studies show that, under the tested Qwen2.5-VL-7B prompts, direct VLM accept/rollback decisions are highly prompt-sensitive, whereas descriptive change-type outputs are structured and useful for semantic auditing but remain context- and scale-sensitive. When those labels are inserted into a hand-crafted per-category gate on the full SYSU-CD test set, they do not outperform a simple probability-only rule (ΔF1=−0.0004, p=0.672). These results motivate the decision–diagnosis separation principle used in this work: let the VLM diagnose and explain, but keep the automatic deployment decision in a transparent arbitration rule.

Guided by this principle, we proposed V4-Safe, a training-free rollback gate that compares the baseline prediction with a TTA-based V3 proposal and rolls back edits whose candidate edit support lies in a high-confidence baseline region. Across the complete 15-cell matrix, V4-Safe gives positive ΔF1 in 13 cells, an exact no-op in the degenerate SADE-NJDS cell, and a statistically indistinguishable −0.000001 change in SNUNet-LEVIR, with significant gains in 9 cells. It reduces harmful edits substantially on the in-house SYSU-CD cells, including 41.90% to 7.68% on PDEFM-SYSU. BIT-NJDS reduces harmful edit rate from 10.87% to 4.88%, while the public-backbone SYSU-CD cells reveal support-score scale drift under the same global threshold. We therefore view V4-Safe as a transparent expert-system arbitration layer and a useful baseline for future gates, not as a final universal threshold. The added SYSU validation-split audit further indicates that the operating threshold should be selected on a target validation split when the checkpoint or data distribution changes, rather than assumed to transfer unchanged.

The broader message is methodological: CD verification should be evaluated not only by mean F1, but also by harmful edit rate, Oracle gap, and stratum-level failure profiles. Future work should add domain-specific or quantile-based threshold calibration, learned lightweight arbitration for residual low-confidence failures, multi-VLM comparisons, and external-backbone evaluation on additional datasets.

CRediT authorship contribution statement

Zhenyu Wang: Conceptualization, Methodology, Software, Investigation, Writing – original draft; **Gang Shi:** Supervision, Writing – review & editing, Resources, Funding acquisition.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used GPT-5 and GPT-5 Proin order to improve the readability and language of selected sections of the manuscript. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the Key Research and Development Project in Xinjiang Uygur Autonomous Region (No. 2022B01006).

Data availability

The datasets used in this study (SYSU-CD, LEVIR-CD, NJDS) are publicly available. The code is available from the corresponding author upon reasonable request.

Appendix A. Statistical robustness analysis

To complement the Wilcoxon p-values reported in Table 5, we provide a bootstrap-based assessment of effect-size stability and report ϵ-harmful frequencies that quantify sensitivity of the harmful-edit metric to small per-sample F1 perturbations.

For each complete correction-artifact configuration we draw 10,000 paired bootstrap resamples of the per-sample F1 difference $d^{(k)} = \mathrm{F1}^{(k)}_{\text{V4-Safe}} - \mathrm{F1}^{(k)}_{\text{baseline}}$ and report the 95% percentile interval for mean ΔF1. The resampling seeds are 20,260,706 for the original 11 cells and 20,260,712 for the four subsequently recovered BIT/SNUNet LEVIR-CD/NJDS cells. We additionally compute all-sample ϵ-harmful frequencies,

$$\mathrm{Harmful}_{\epsilon}(D) = \frac{1}{N}\sum_{k=1}^{N} \mathbb{1}\left[\mathrm{F1}(\hat{y}^{(k)}_{\text{out}}, y^{(k)}) - \mathrm{F1}(\hat{y}^{(k)}_{\text{b}}, y^{(k)}) < -\epsilon\right], \tag{A.1}$$

for $\epsilon \in \{0, 0.001, 0.005, 0.01\}$, which controls for sub-threshold F1 fluctuations. Unlike the primary harmful edit rate in Eq. (5), this sensitivity frequency uses all N samples as its denominator; at $\epsilon = 0.005$, its numerator matches the primary metric's numerator. Finally, the Wilcoxon p-values from Table 5 are corrected for multiple comparisons across the complete cells using the Holm–Bonferroni procedure (Table A.1).

Interpretation. The bootstrap intervals broadly track the Wilcoxon significance pattern in Table 5. Nine cells remain significant after Holm correction. The BIT-NJDS percentile CI is positive despite a non-significant Wilcoxon test; this difference reflects the distinct null hypotheses of a mean bootstrap interval and a signed-rank test, so we retain the more conservative n.s. label in the main table. The daggered SADE-NJDS row is a no-edit degenerate cell with zero delta, consistent with the saturated/degenerate-regime explanation in Section 5.2. The ϵ sensitivity analysis shows how many negative per-sample changes remain after progressively excluding small F1 fluctuations. For example, the SADE-SYSU all-sample frequency falls from 20.28% at strict decrease to 9.28% at the primary 0.005 margin. This distinction is why the main tables report the edit-conditioned ϵ=0.005 metric and this appendix exposes all four margins. Holm correction does not change the qualitative pattern of 9 significant cells.

Table A.1
Bootstrap-based 95% CI for ΔF1 (V4-Safe vs baseline), all-sample ϵ-harmful frequencies, and Holm-corrected p-values across the complete correction-artifact configurations. The original Wilcoxon p-values appear in Table 5.

Model	Dataset	N	ΔF1	95% CI	H_0	$H_{0.001}$	$H_{0.005}$	$H_{0.01}$	p_{Holm}
SADE	SYSU	4000	0.0051	[0.0040, 0.0063]	20.28%	16.10%	9.28%	5.83%	7.04e-26
SADE	LEVIR	128	0.0012	[0.0005, 0.0020]	8.59%	3.13%	0.00%	0.00%	1.03e-4
SADE	NJDS†	504	0.0000	[0.0000, 0.0000]	0.00%	0.00%	0.00%	0.00%	n.a.
PDEFM	SYSU	3999	0.0010	[0.0007, 0.0013]	4.13%	2.35%	1.18%	0.65%	3.10e-31
PDEFM	LEVIR	128	0.0004	[0.0002, 0.0006]	1.56%	0.78%	0.00%	0.00%	1.97e-4
PDEFM	NJDS	1008	0.0004	[-0.0002, 0.0010]	1.39%	1.39%	1.19%	0.99%	5.30e-1
DCCA	SYSU	3991	0.0015	[0.0011, 0.0018]	4.66%	2.63%	1.35%	0.93%	2.62e-34
DCCA	LEVIR	128	0.0006	[0.0002, 0.0013]	2.34%	0.78%	0.00%	0.00%	8.40e-3
DCCA	NJDS	1008	0.0001	[-0.0006, 0.0007]	1.98%	1.79%	1.59%	1.29%	6.84e-1
BIT	SYSU	4000	0.0039	[0.0026, 0.0052]	27.10%	23.53%	17.55%	12.83%	1.52e-32
BIT	LEVIR	128	0.0008	[0.0003, 0.0016]	9.38%	2.34%	0.00%	0.00%	1.88e-6
BIT	NJDS	504	0.0034	[0.0007, 0.0081]	0.60%	0.40%	0.40%	0.00%	5.30e-1
SNUNet	SYSU	4000	0.0072	[0.0060, 0.0085]	28.98%	25.95%	19.53%	15.08%	2.08e-33
SNUNet	LEVIR	128	-0.000001	[-0.000010, 0.000005]	0.78%	0.00%	0.00%	0.00%	6.84e-1
SNUNet	NJDS	504	0.0002	[0.00008, 0.00042]	0.40%	0.20%	0.00%	0.00%	6.84e-1

Appendix B. VLM diagnostic protocol and prompt reproducibility

Spatial input and coordinate mapping. Disputed patches are indexed by their original integer top-left coordinate (y, x) and use a 64×64 square window. The same coordinate slice is applied to T1, T2, and every prediction/mask panel, preserving one-to-one pixel alignment and aspect ratio. The base 256×256 inputs are divisible by 64, so the base grid does not require border padding. The 128- and 192-pixel context views are centered on the same patch and are used only for the diagnostic context ablation. No VLM image is resized back into a prediction map, and the deployment-time gate always uses the full-resolution probability and edit support.

Decision-prompt instructions. V1–V3 append the following instruction to a common task preamble and the aligned T1/T2/baseline/V3 panels: V1, "Should the proposed edit be kept or suppressed?"; V2, the same binary alternatives in reversed order; and V3, "Keep / Suppress / Uncertain." V4 is the archived three-image pilot instruction "Score the reliability of the proposed edit in $[0, 1]$." Because V4 differs in both instruction and panel count, V2 versus V3 on 532 identical patches is the strict prompt-only comparison; V4 is reported as an auxiliary collapse case.

Exact V5 open-description prompt. The 7B, 32B-AWQ, and 72B-AWQ runs use the same user message:

```
You are analyzing a satellite image pair
for change detection.
Image 1: satellite image at time T1.
Image 2: satellite image at time T2.
Image 3: the model's predicted change mask
(white = predicted change).
Describe what you observe. Answer these
three questions:
1. change_type: what kind of change do you
see?
Choose one: building, road, vegetation,
water, demolition, shadow, noise, none.
2. confidence: how confident are you?
Choose one: high, medium, low.
3. description: one sentence describing
what changed or why nothing changed.
Output EXACTLY in this format:
change_type: <type>
confidence: <level>
description: <sentence>
```

Generation is deterministic (`do_sample=False`, `max_new_tokens=128`); images are supplied as native square RGB PNGs to the Qwen AutoProcessor. Each retained disputed patch triggers one generation call. V1 uses 1000 pilot calls; V2 and V3 use 532 calls each over the same 100 samples; V4 uses 182 calls over 30 samples. The common-100 32B/72B scale check uses 532 calls per model, and each 500-sample scale check uses 2395 calls per model. The full 7B V5 run aggregates all retained patch calls into 3675 valid sample-level diagnoses among the 4000 SYSU test samples.

Paired qualitative example. For SYSU sample 00003 at patch coordinate `y128_x192`, the same-input decision prompts disagree: V2 returns keep, V3 returns uncertain, and the auxiliary V4 probe returns suppress. In descriptive mode, the 72B model assigns road with high confidence across the sample's patches, repeatedly describing a new, widened, or altered road. Thus, the semantic description is coherent while the requested direct action changes with the decision prompt, illustrating why the description is retained for diagnosis but excluded from automatic control.

References

Angelopoulos, A. N., & Bates, S. (2023). Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, *16*(4), 494–591. https://doi.org/10.1561/2200000101

Angelopoulos, A. N., Kohli, A. P., Bates, S., Jordan, M. I., Malik, J., Alshaabi, T., & Upadhyayula, S. (2022). Image-to-image regression with distribution-free uncertainty quantification and applications in imaging. In *Proceedings of the 39th international conference on machine learning* (pp. 717–730). (*vol. 162*). Proceedings of Machine Learning Research.

Bai, S., Chen, K., Liu, X., Wang, J., Ge, W., Song, S., Dang, K., Wang, P., Wang, S., Tang, J., Zhong, H., Zhu, Y., Yang, M., Li, Z., Wan, J., Wang, P., Ding, W., Fu, Z., Xu, Y., Ye, J., Zhang, X., Xie, T., Cheng, Z., Zhang, H., Yang, Z., Xu, H., & Lin, J. (2025). Qwen2.5-VL technical report. https://doi.org/10.48550/arXiv.2502.13923

Bandara, W. G. C., & Patel, V. M. (2022). A transformer-based Siamese network for change detection. In *IGARSS 2022–2022 IEEE international geoscience and remote sensing symposium* (pp. 207–210). https://doi.org/10.1109/IGARSS46834.2022.9883686

Chen, H., Qi, Z., & Shi, Z. (2022). Remote sensing image change detection with transformers. *IEEE Transactions on Geoscience and Remote Sensing*, *60*, 1–14. https://doi.org/10.1109/TGRS.2021.3095166

Chen, H., & Shi, Z. (2020). A spatial-temporal attention-based method and a new dataset for remote sensing image change detection. *Remote Sensing*, *12*(10), 1662. https://doi.org/10.3390/rs12101662

Chen, K., Liu, C., Chen, H., Zhang, H., Li, W., Zou, Z., & Shi, Z. (2024). RSPrompter: Learning to prompt for remote sensing instance segmentation based on visual foundation model. *IEEE Transactions on Geoscience and Remote Sensing*, *62*, 1–17. https://doi.org/10.1109/TGRS.2024.3356074

Daudt, R. C., Le Saux, B., & Boulch, A. (2018). Fully convolutional siamese networks for change detection. In *2018 25th IEEE international conference on image processing (ICIP)* (pp. 4063–4067). https://doi.org/10.1109/ICIP.2018.8451652

Fang, S., Li, K., Shao, J., & Li, Z. (2022). SNUNet-CD: A densely connected Siamese network for change detection of VHR images. *IEEE Geoscience and Remote Sensing Letters*, *19*, 1–5. https://doi.org/10.1109/LGRS.2021.3056416

Fu, Y., Ni, A., Peng, H., & Peng, J. (2026). AFSNet: Remote sensing change detection based on progressive difference feature aggregation and full-scale connection. *Expert Systems with Applications*, *292*, 129994. https://doi.org/10.1016/j.eswa.2025.129994

Gao, J., Zhang, D., Wang, F., Ning, L., Zhao, Z., & Li, X. (2025). Combining SAM with limited data for change detection in remote sensing. *IEEE Transactions on Geoscience and Remote Sensing*, *63*, 1–11. https://doi.org/10.1109/TGRS.2025.3545040

Gao, Y., Pei, G., Sheng, M., Sun, Z., Chen, T., & Yao, Y. (2024). Relating CNN-Transformer fusion network for remote sensing change detection. In *2024 IEEE International conference on multimedia and expo (ICME)* (pp. 1–6). https://doi.org/10.1109/ICME57554.2024.10687791

Gawlikowski, J., Tassi, C. R. N., Ali, M., Lee, J., Humt, M., Feng, J., Kruspe, A., Triebel, R., Jung, P., Roscher, R., Shahzad, M., Yang, W., Bamler, R., & Zhu, X. X. (2023). A survey of uncertainty in deep neural networks. *Artificial Intelligence Review*, *56*(Suppl 1), 1513–1589. https://doi.org/10.1007/s10462-023-10562-9

Geifman, Y., & El-Yaniv, R. (2017). Selective classification for deep neural networks. In *Advances in neural information processing systems* (pp. 4885–4894). (*vol. 30*).

Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. In *Proceedings of the 34th international conference on machine learning* (pp. 1321–1330). (*vol. 70*). Proceedings of Machine Learning Research.

Hu, Y., Yuan, J., Wen, C., Lu, X., Liu, Y., & Li, X. (2025). RSGPT: A remote sensing vision language model and benchmark. *ISPRS Journal of Photogrammetry and Remote Sensing*, *224*, 272–286. https://doi.org/10.1016/j.isprsjprs.2025.03.028

Jiang, H., Kim, B., Guan, M. Y., & Gupta, M. (2018). To trust or not to trust a classifier. In *Advances in neural information processing systems* (pp. 5546–5557). (*vol. 31*).

Kuckreja, K., Danish, M. S., Naseer, M., Das, A., Khan, S., & Khan, F. S. (2024). Geochat: Grounded large vision-language model for remote sensing. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 27831–27840). https://doi.org/10.1109/CVPR52733.2024.02629

Lakshminarayanan, B., Pritzel, A., & Blundell, C. (2017). Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in neural information processing systems* (pp. 6402–6413). (*vol. 30*).

Li, X., Wen, C., Hu, Y., Yuan, Z., & Zhu, X. X. (2024). Vision-language models in remote sensing: Current progress and future trends. *IEEE Geoscience and Remote Sensing Magazine*, *12*(2), 32–66. https://doi.org/10.1109/MGRS.2024.3383473

Liu, C., Chen, K., Zhang, H., Qi, Z., Zou, Z., & Shi, Z. (2024a). Change-Agent: Toward interactive comprehensive remote sensing change interpretation and analysis. *IEEE Transactions on Geoscience and Remote Sensing*, *62*, 1–16. https://doi.org/10.1109/TGRS.2024.3425815

Liu, F., Chen, D., Guan, Z., Zhou, X., Zhu, J., Ye, Q., Fu, L., & Zhou, J. (2024b). RemoteCLIP: A vision language foundation model for remote sensing. *IEEE Transactions on Geoscience and Remote Sensing*, *62*, 1–16. https://doi.org/10.1109/TGRS.2024.3390838

Ning, L., Zhou, Q., Wang, Q., Gao, J., & Li, X. (2025). Cross-resolution change detection in remote sensing via unequal relationships from a frequency perspective. *IEEE Transactions on Geoscience and Remote Sensing*, *63*, 1–14. https://doi.org/10.1109/TGRS.2025.3590023

Osco, L. P., Wu, Q., de Lemos, E. L., Goncalves, W. N., Ramos, A. P. M., Li, J., & Marcato Junior, J. (2023). The segment anything model (SAM) for remote sensing applications: From zero to one shot. *International Journal of Applied Earth Observation and Geoinformation*, *124*, 103540. https://doi.org/10.1016/j.jag.2023.103540

Ovadia, Y., Fertig, E., Ren, J., Nado, Z., Sculley, D., Nowozin, S., Dillon, J. V., Lakshminarayanan, B., & Snoek, J. (2019). Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift. In *Advances in neural information processing systems* (pp. 13969–13980). (*vol. 32*).

Peng, Q., Liu, K., Zou, B., & Li, M. (2025). EfficientSAM-assisted network with feature interaction and priori guidance for remote sensing change detection. *IEEE Geoscience and Remote Sensing Letters*, *22*, 1–5. https://doi.org/10.1109/LGRS.2025.3587852

Shen, Q., Liu, J., Huang, J., Chen, X., & Zhang, L. (2022). Semantic feature-constrained multitask siamese network for building change detection in high-resolution remote sensing images. *ISPRS Journal of Photogrammetry and Remote Sensing*, *189*, 78–94. https://doi.org/10.1016/j.isprsjprs.2022.04.015

Shi, Q., Liu, M., Li, S., Liu, X., Wang, F., & Zhang, L. (2022). A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection. *IEEE Transactions on Geoscience and Remote Sensing*, *60*, 1–16. https://doi.org/10.1109/TGRS.2021.3085870

Sun, Y., Wang, X., Liu, Z., Miller, J., Efros, A. A., & Hardt, M. (2020). Test-time training with self-supervision for generalization under distribution shifts. In *Proceedings of the 37th international conference on machine learning* (pp. 9229–9248). (*vol. 119*). Proceedings of Machine Learning Research.

Wang, D., Shelhamer, E., Liu, S., Olshausen, B., & Darrell, T. (2021). Tent: Fully test-time adaptation by entropy minimization. In *International conference on learning representations*. https://openreview.net/forum?id=uXl3bZLkr3c.

Wang, Z., Wang, G., & Shi, G. (2026a). PDEFM-Net: Progressive difference enhancement and deep feature mining with spatial sequence modeling for remote sensing change detection. In *Proceedings of the international conference on intelligent computing (ICIC)* (p. to appear). Accepted, to appear.

Wang, Z., Wang, G., & Shi, G. (2026b). SADE-Net: Strip-aware difference enhancement and multi-dilation fusion for remote sensing change detection. In *Proceedings of the international joint conference on neural networks (IJCNN)* (p. to appear). Accepted, to appear.

Wang, Z., Yang, X., & Shi, G. (2026c). DCCA-Net: Difference-driven coordinate attention and strip context network for remote sensing change detection. In *Proceedings of the international conference on intelligent computing (ICIC)* (p. to appear). Accepted, to appear.

Wilcoxon, F. (1945). Individual comparisons by ranking methods. *Biometrics Bulletin*, *1*(6), 80–83. https://doi.org/10.2307/3001968

Yang, H., Ren, B., Yang, Z., Xiong, J., Li, X., & Chen, C. Y.-C. (2026). HWA-Net: Hierarchical window aggregate network for remote sensing image change detection. *Expert Systems with Applications*, *292*, 129829. https://doi.org/10.1016/j.eswa.2025.129829

Yuan, Q., & Du, Z. (2026). DCIBCD: A dual-branch cooperative interaction method for interference-resistant binary change detection. *Expert Systems with Applications*, *301*, 130210. https://doi.org/10.1016/j.eswa.2025.130210

Zhang, D., Wang, F., Ning, L., Zhao, Z., Gao, J., & Li, X. (2024a). Integrating SAM with feature interaction for remote sensing change detection. *IEEE Transactions on Geoscience and Remote Sensing*, *62*, 1–11. https://doi.org/10.1109/TGRS.2024.3483775

Zhang, H., Chen, K., Liu, C., Chen, H., Zou, Z., & Shi, Z. (2025). CDMamba: Incorporating local clues into mamba for remote sensing image binary change detection. *IEEE Transactions on Geoscience and Remote Sensing*, *63*, 1–16. https://doi.org/10.1109/TGRS.2025.3545012

Zhang, M., Levine, S., & Finn, C. (2022). MEMO: Test time robustness via adaptation and augmentation. In *Advances in neural information processing systems* (pp. 38629–38642). (*vol. 35*).

Zhang, W., Cai, M., Zhang, T., Zhuang, Y., & Mao, X. (2024b). EarthGPT: A universal multi-modal large language model for multi-sensor image comprehension in remote sensing domain. *IEEE Transactions on Geoscience and Remote Sensing*, *62*, 1–20. https://doi.org/10.1109/TGRS.2024.3409624

Zheng, Z., Zhong, Y., Tian, S., Ma, A., & Zhang, L. (2022). Changemask: Deep multi-task encoder-transformer-decoder architecture for semantic change detection. *ISPRS Journal of Photogrammetry and Remote Sensing*, *183*, 228–239. https://doi.org/10.1016/j.isprsjprs.2021.10.015